Section 7 Mechanical installation (Part 2)

Contents

1. INTRODUCTION

This section describes the mechanical installation of the following units:

Figure 1 – Magnet right side view

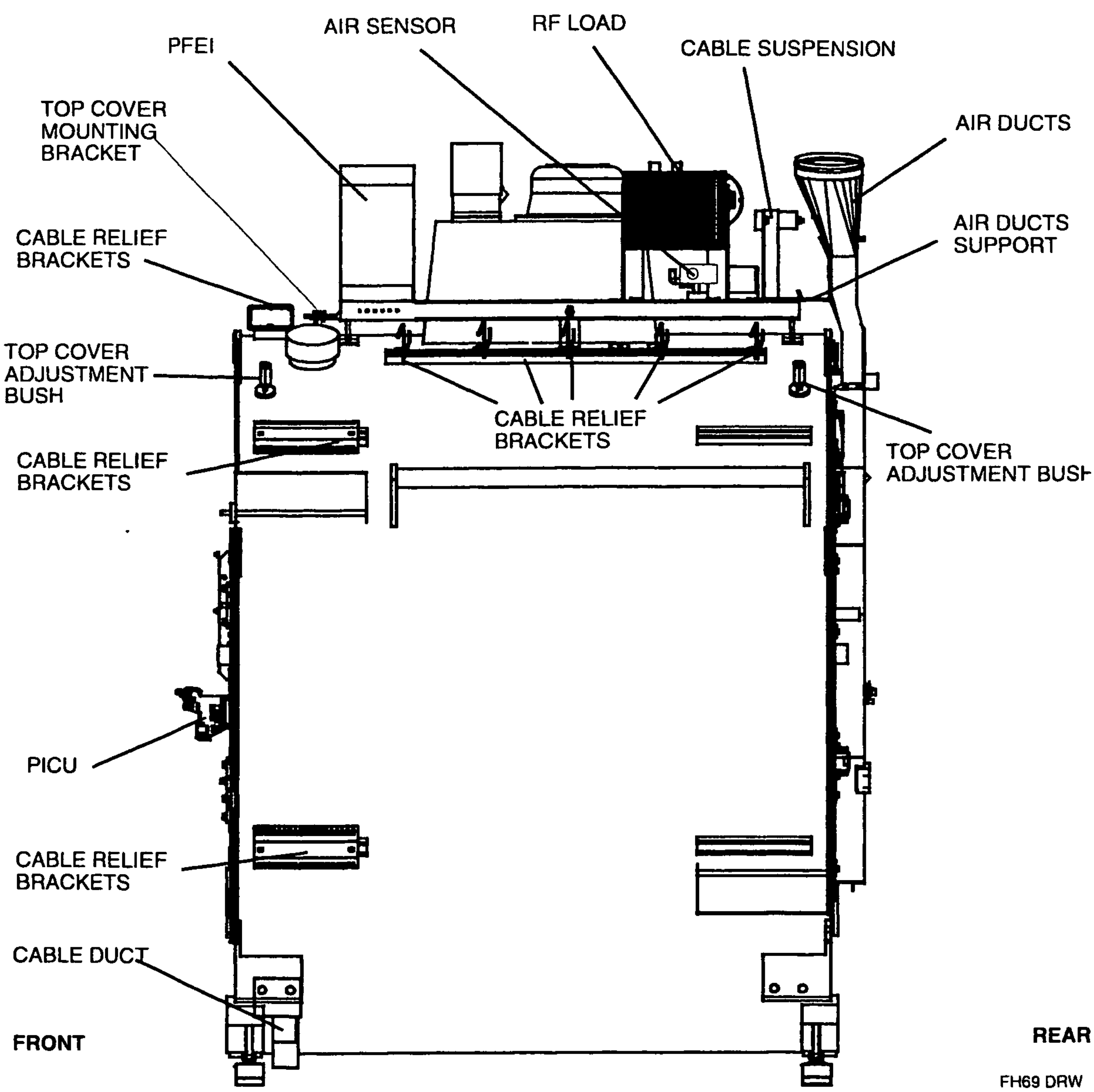

Figure 2 – Magnet rear view

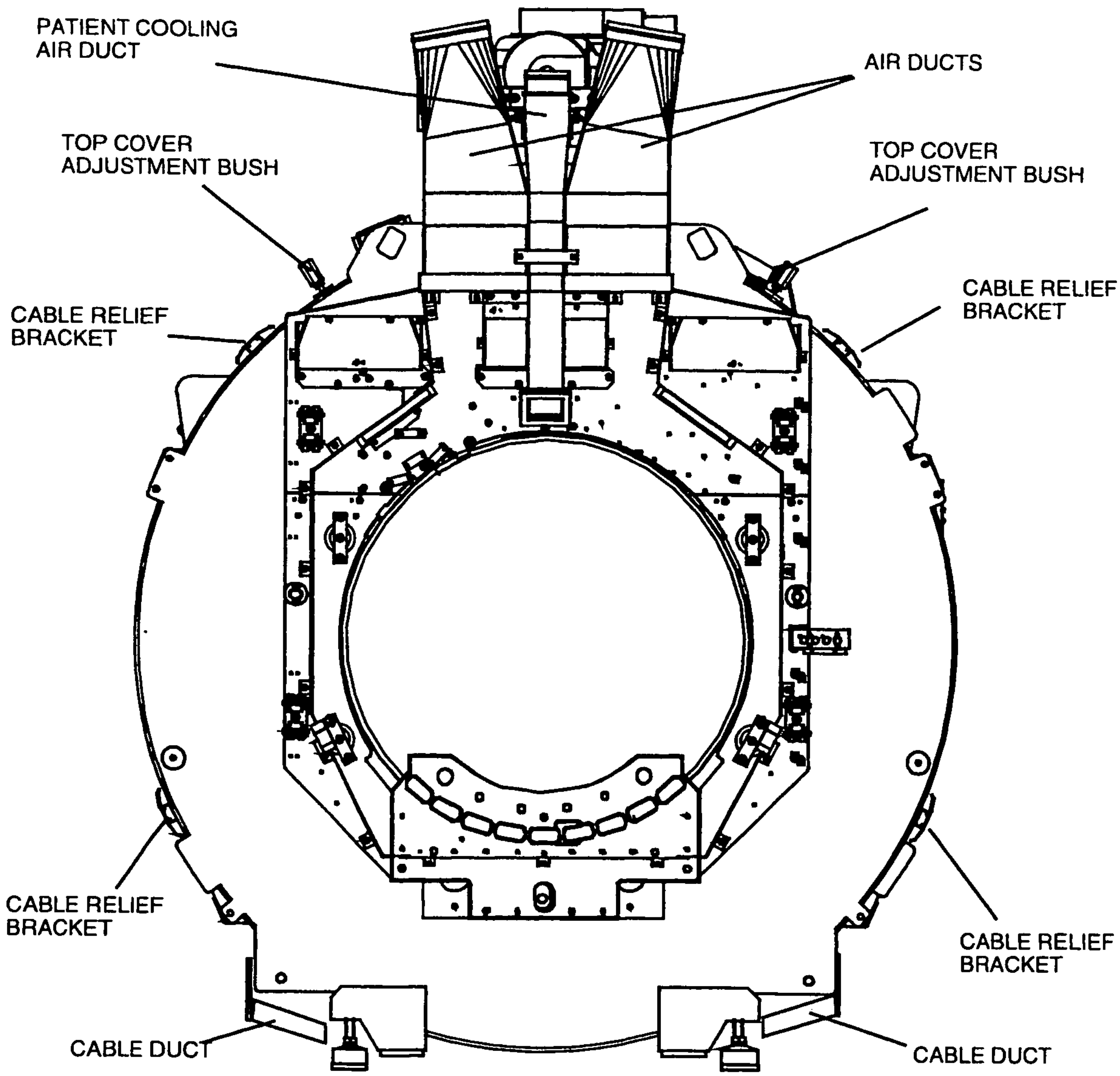

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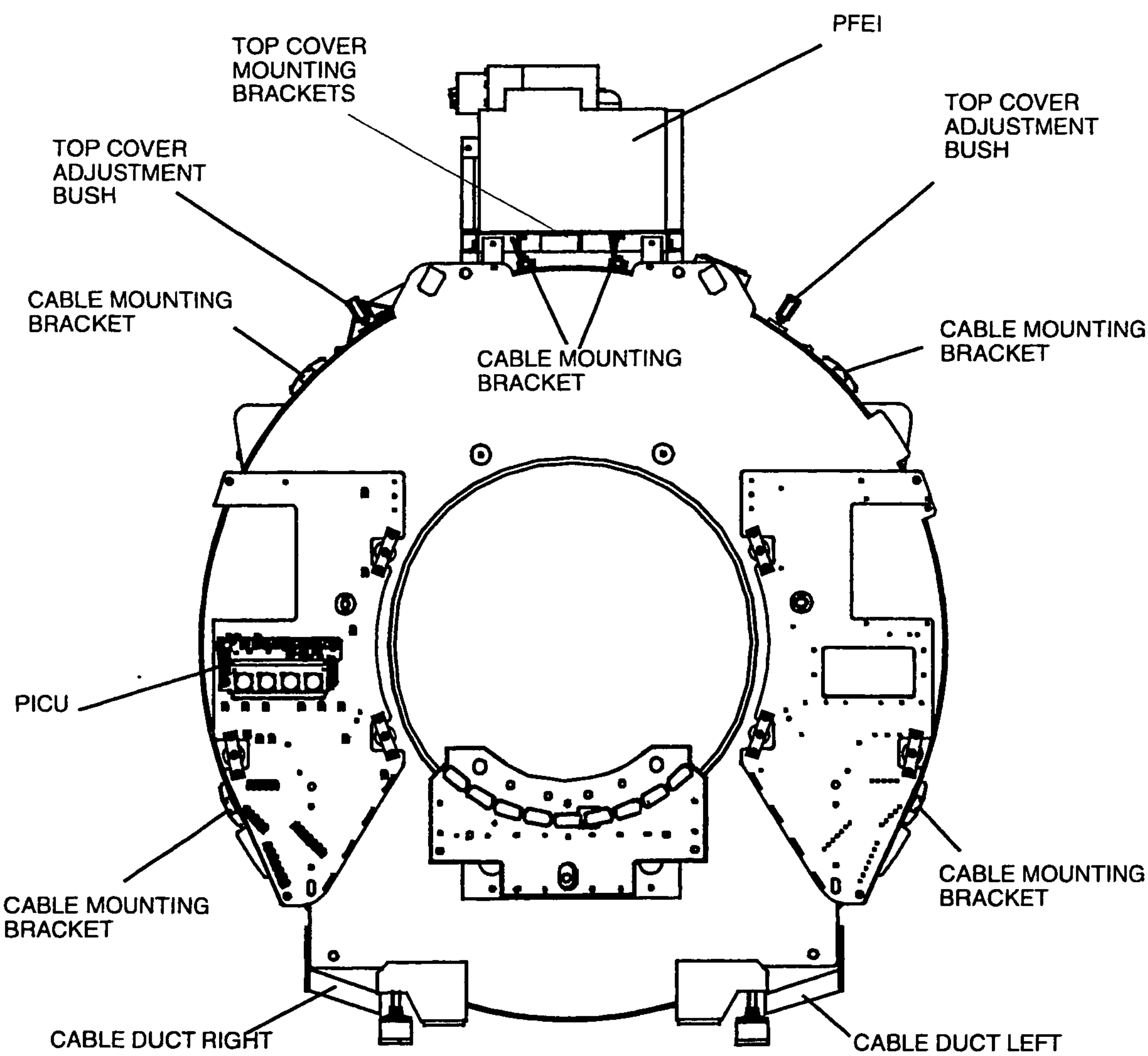
Figure 3 – Magnet Front view



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2. MOUNTING THE PFEI

Mount the PFEI on top of the magnet top-frame at the front side of the magnet.
The service key stickers on the PFEI must point towards the patient end.

Figure 4 – Position the PFEI

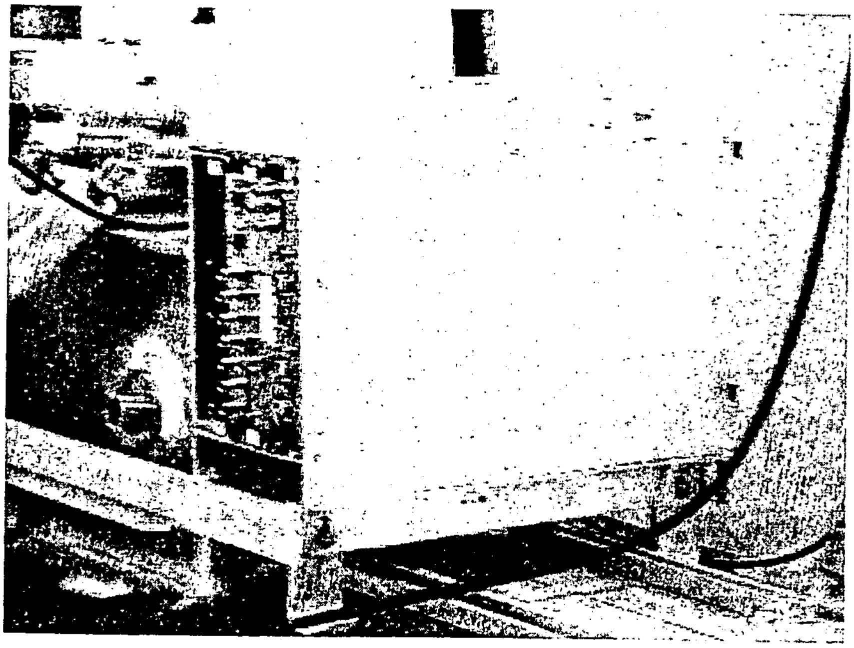
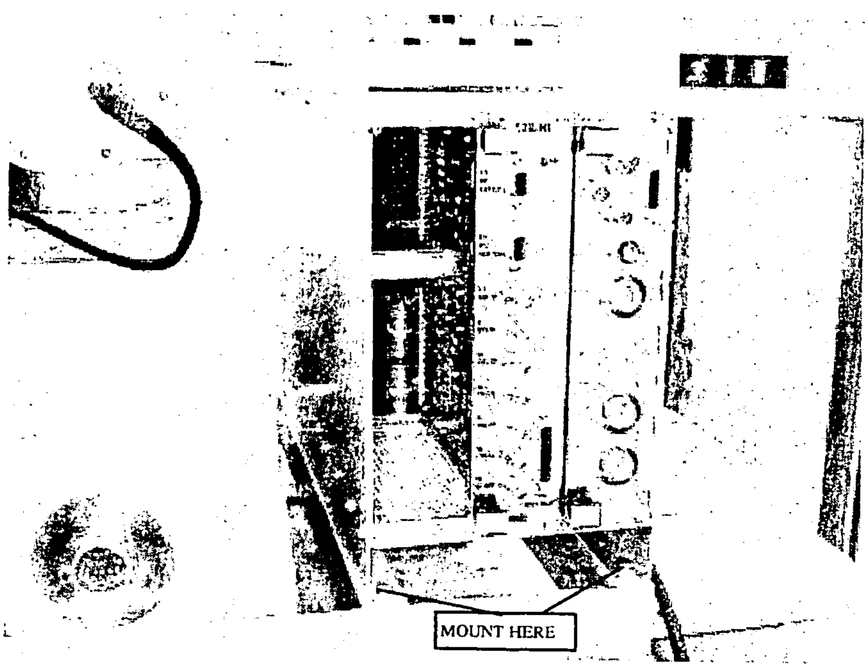
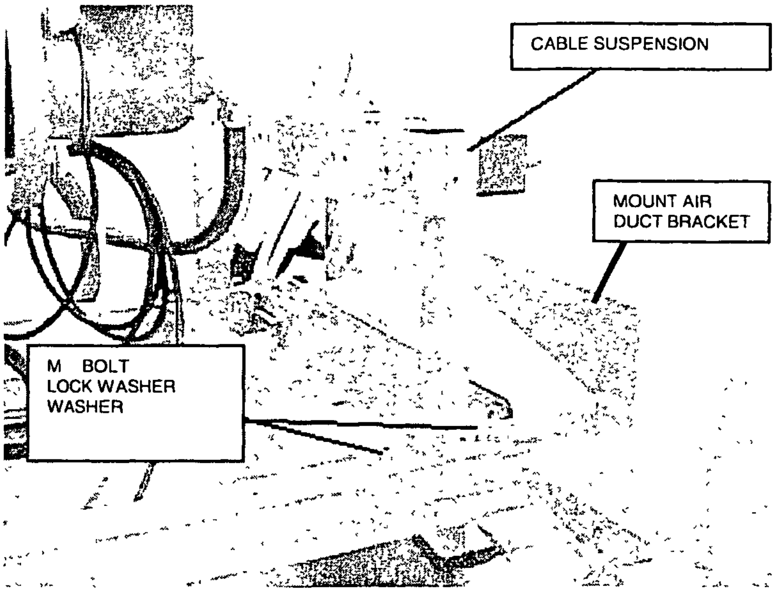


Figure 5 – Mount the PFEI



3. MOUNTING THE CABLE SUSPENSION

Mount the cable suspension on top of the magnet top-frame at the rear side of the magnet.

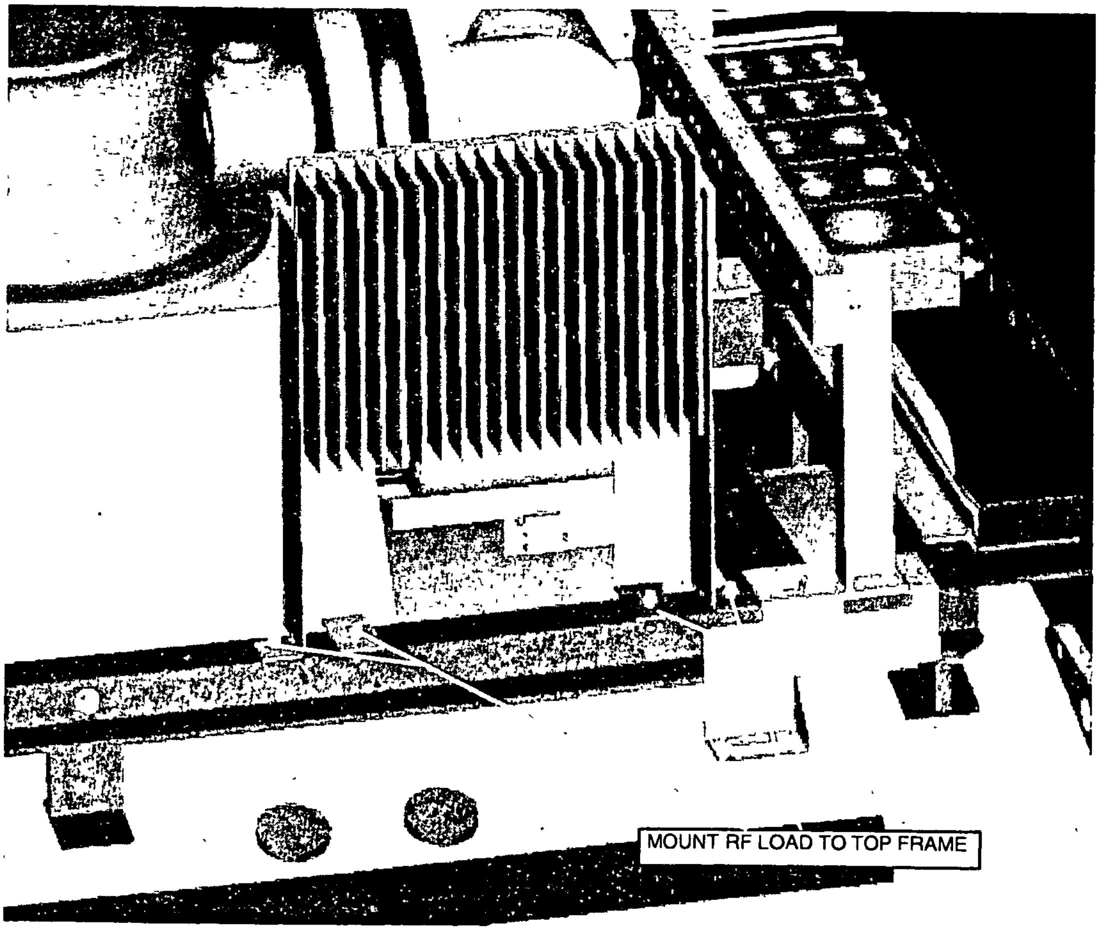


4. MOUNT THE RF LOAD ASSY

Mount the RF load assy on top of the magnet top-frame at the magnet right/rear side.

Mount the RF load with the RF load connector towards the center of the magnet otherwise the top covers will not fit.

Figure 6 – Mounting the RF load



5. MOUNTING THE CABLE RELEAF BRACKETS

Four cable relief brackets must be mounted to the magnet. Two brackets at the left side and two at the right side.

Figure 7 – cable bracket (left bottom side)

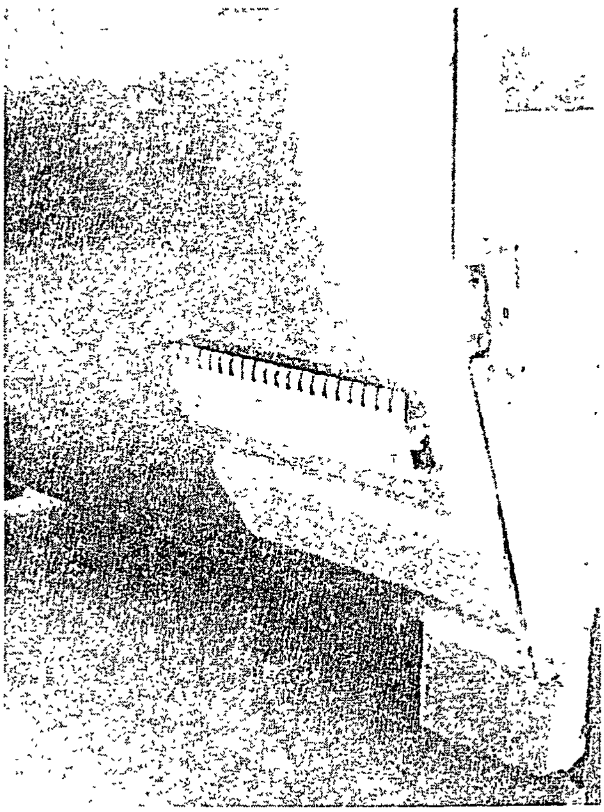
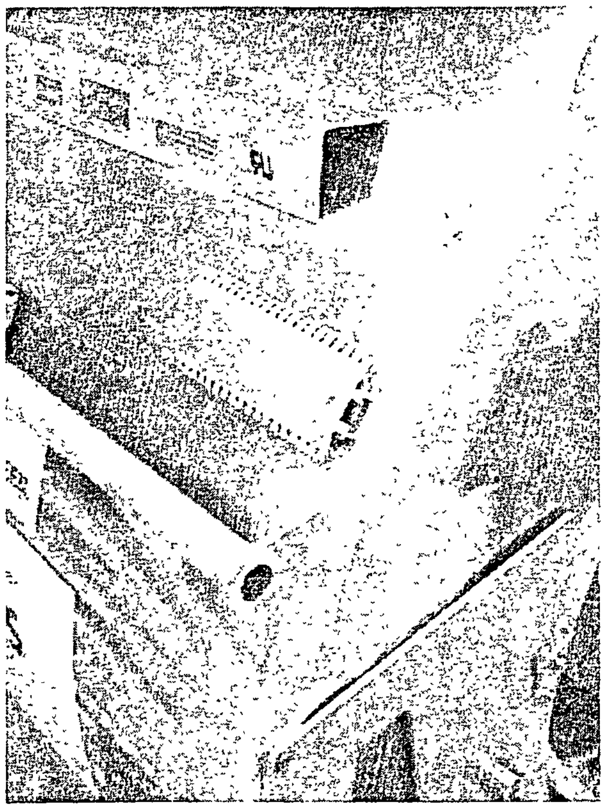
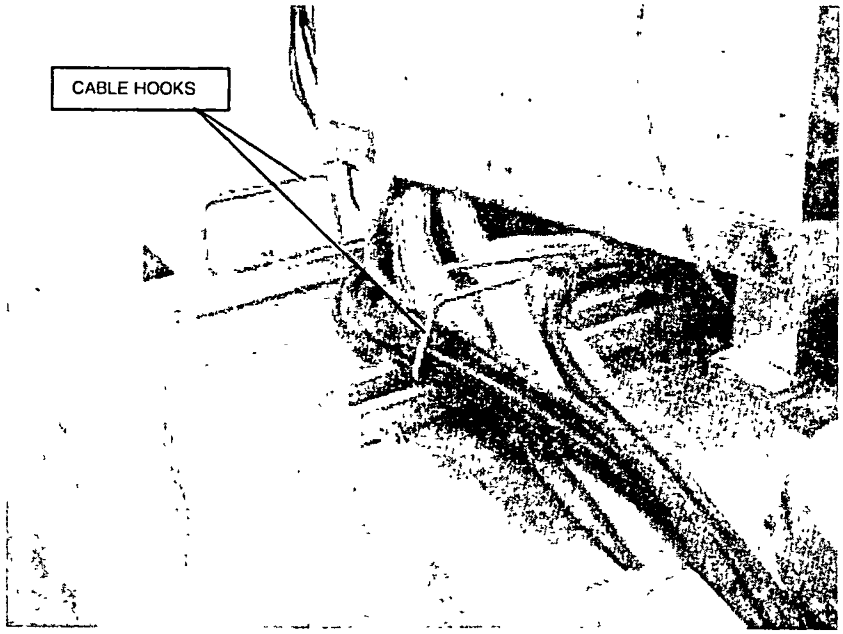


Figure 8 – cable bracket (left top side)



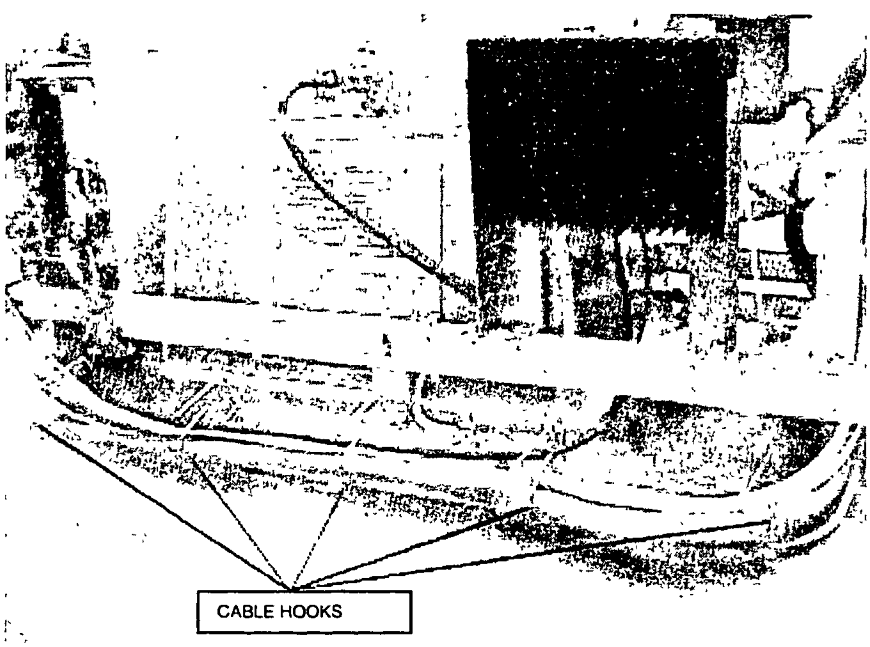
Mount two cable hooks at the top front side of the magnet (before the PFEI).

Figure 9 – Mount cable hooks



Mount four cable hooks to the magnet at the right topside.

Figure 10 – Mount cable hooks



6. MOUNTING THE TOPCOVER BRACKETS

1. Mount the top-cover bracket at the front of the magnet against the magnet top frame.

Figure 11 – bracket at magnet front

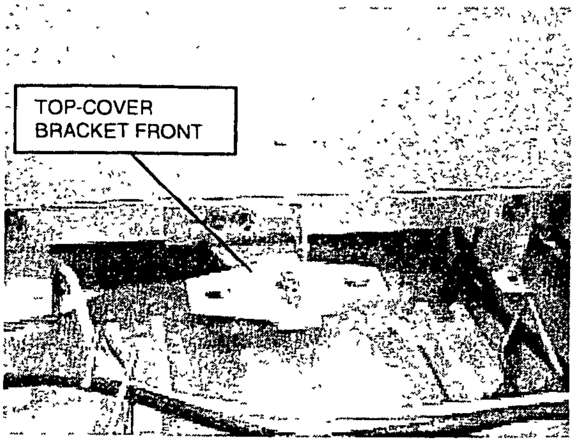
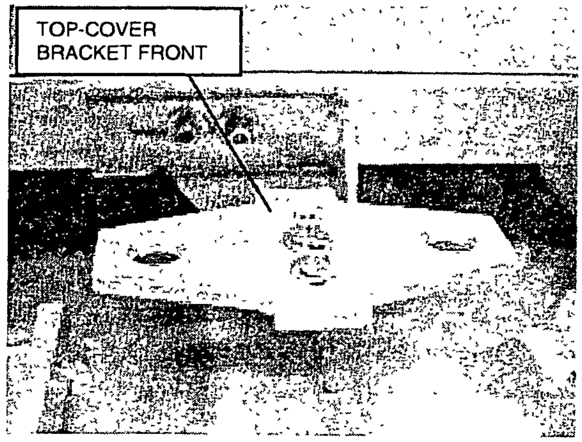


Figure 12 - bracket at magnet front



2. Mount two top cover brackets together with the cable suspension

Figure 13 – Top cover brackets

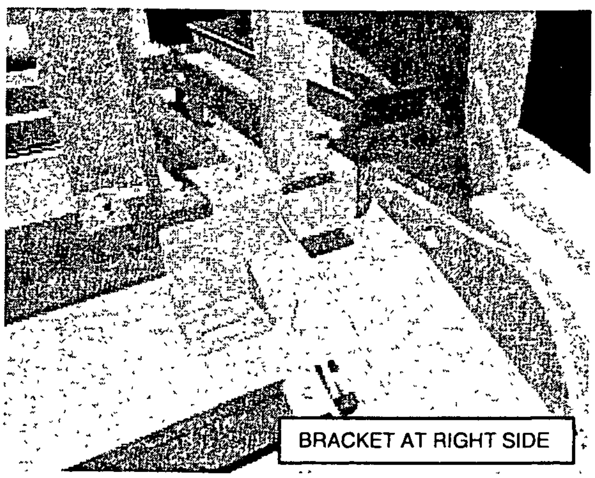
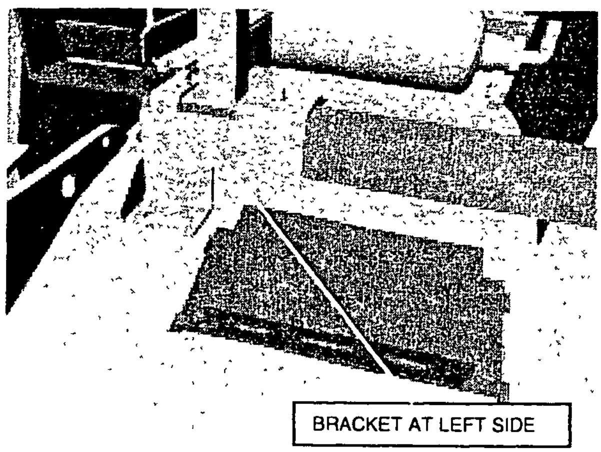
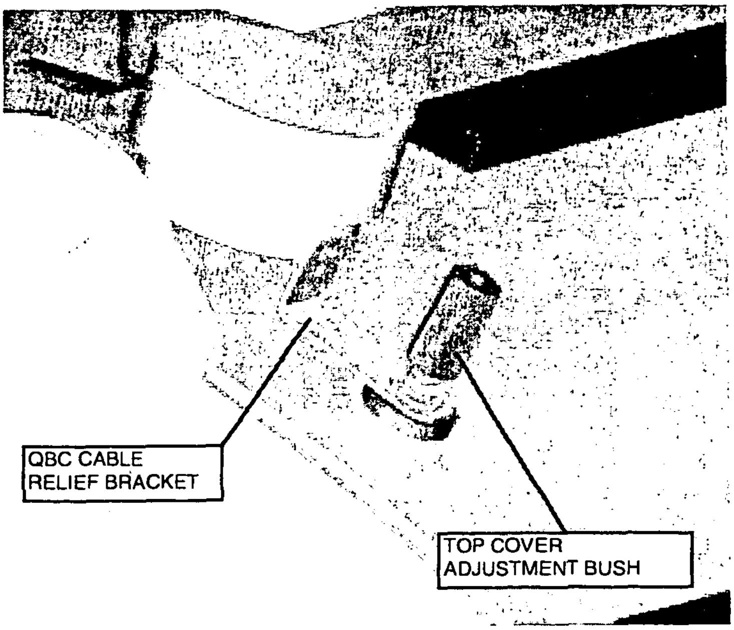


Figure 14 – Top cover brackets



3. Mount four top cover adjustment bushes on top of the magnet.
   At the front right side of the magnet, mount the "QBC cable relief bracket". Mount this bracket together with the "top cover adjustment bush".

Figure 15 - Top cover adjustment bush



7. MOUNTING THE CABLE DUCTS

At the left <u>and</u> right side of the magnet, mount a cable duct.
The cables towards the patient support are rooted trough these cable ducts

Figure 16 – Location of cable duct

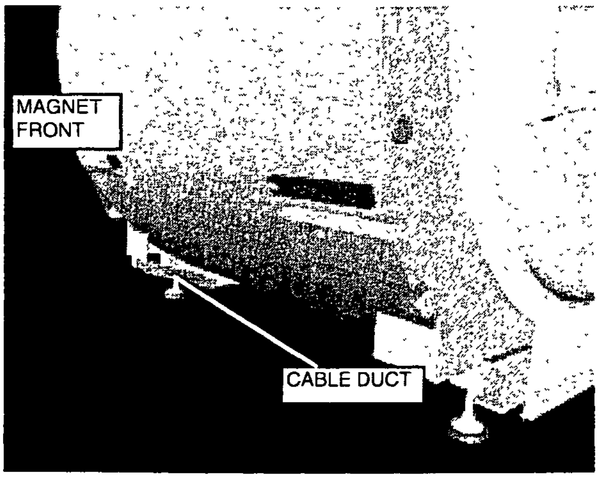
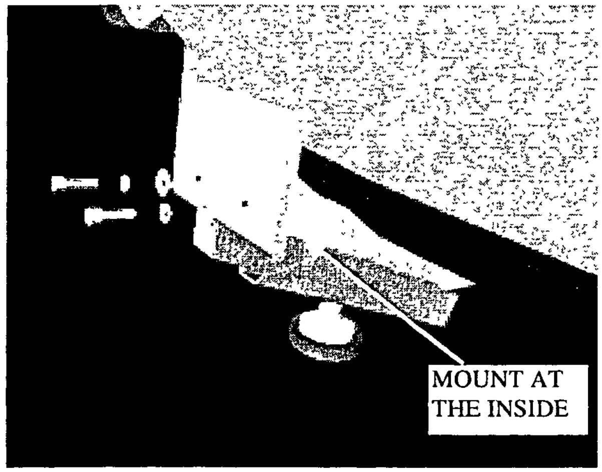


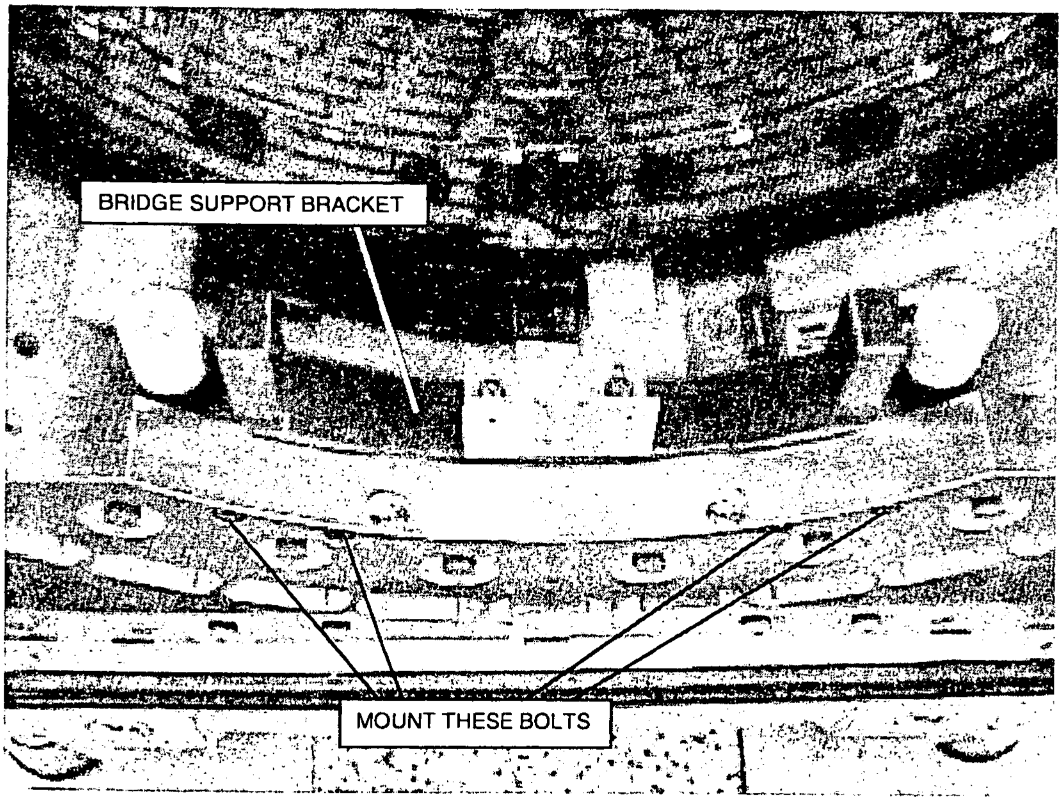
Figure 17 – Mounting the cable duct



8. MOUNTING THE BRIDGE SUPPORT BRACKETS

Mount the front bridge support brackets (bracket with wire for the switch plate) against the front gradient carrier. Mount the other bridge support bracket against the rear gradient carrier.

Figure 18 – Mounting the bridge support bracket



9. MOUNTING THE GRADIENT STRIPS

Skip this paragraph if you have a factory assembled magnet system

9.1. GENERAL

The gradient strips must be mounted on the rear ringframe. For installation details, refer to drawing Z6-4.8 and Z6-4.9 of the drawing manual.

NOTE

Special attention must be paid to ensure that the surfaces are clean and free from dirt and/or resin. Also the fibre glass insulation must not get caught under the cable.
If the insulation needs trimming use a small sharp scalpel. You must wear safety goggles in case the blade breaks.

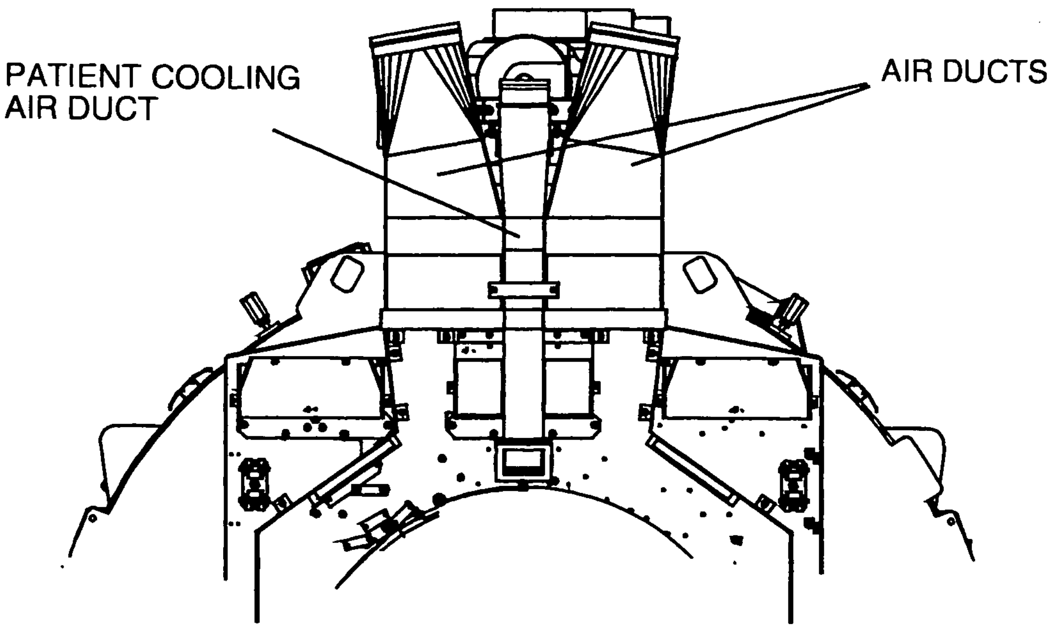
Mount the strips to the gradient coil as follows: Torque the first nut with **40 Nm** using a torque wrench. Apply also 40 Nm to the second nut without touching the first nut.

10. MOUNTING THE AIR DUCTS

Skip this paragraph if you have a factory assembled magnet system

The gradient cables must be mounted before mounting the air ducts. Mount first the left and right air ducts and than the patient cooling air duct.

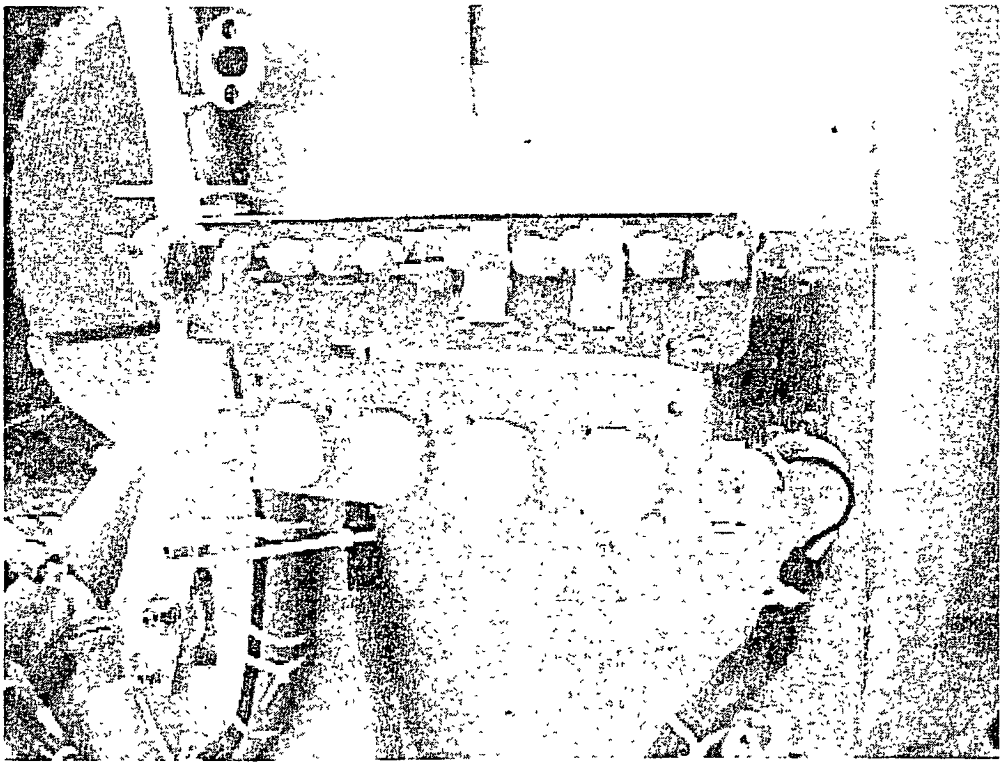
Figure 19 – Air ducts



11. MOUNTING THE PATIENT INTERFACE CONTROL UNIT (PICU)

The PICU can be installed at the left or right side of the front ringframe. This position is site dependent; so ask the hospital staff where they want to have it mounted. Note that the PICU must be at the same side as the Physiology unit in the patient support. Also note that a trolley must be docked at to the patient table at the opposite side of the PICU and Physiology unit.

Figure 20 – Mount the PICU



The rear PICU is already mounted onto the rear ringframe. For factory pre-assembled magnet systems the rear PICU is attached to the rear ringframe with cable binders. In that case cut the cable binders and mount the rear PICU to the rear ringframe.

12. INSTALL THE INTERA PATIENT SUPPORT

NOTE

Before shimming the magnet, the patient support must be positioned in front of the magnet in the upmost position.

12.1. REMOVE TRANSPORT PROVISIONS

To prevent damage to the patient support, wooden wedges are mounted at the right and left side into the patient support.

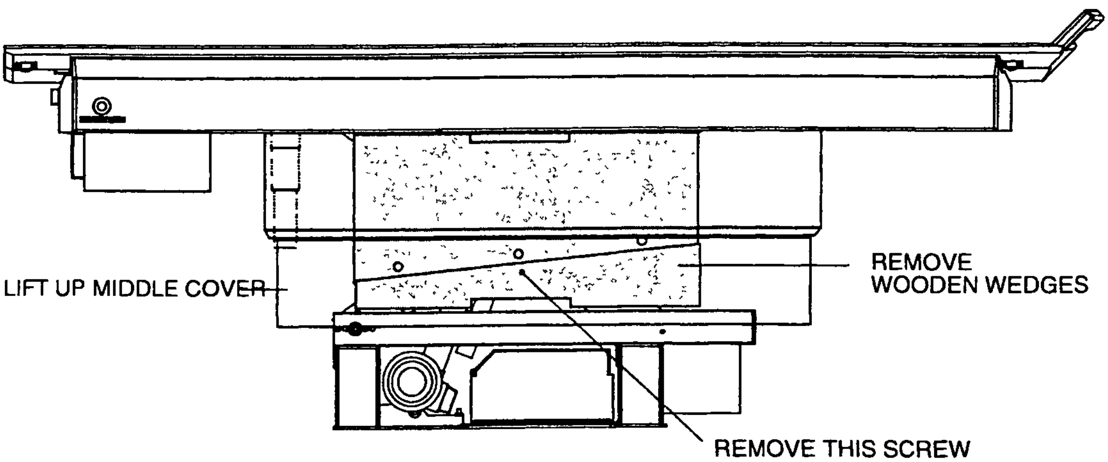
CAUTION

Vertical activation of the patient support, while the transport provisions are installed, will damage the patient support.

Remove the two transport provisions (left and right side) as follows:

1. With an electrical drill unscrew the screws holding the carton box. Remove the carton box.
2. Remove the box with the lower covers, the bag with the insulation spacers and mounting material, and the wooden alignment bars. Store these items in the magnet room. They will be used during the installation of the patient support.
3. Lift up the middle covers. It might be needed to put something underneath the middle covers, this to prevent it from falling down.
4. Take a Phillips screwdriver and a hammer, these are the tools needed to remove the transport provisions.
5. Remove the screws that lock the upper and lower wooden wedges.
6. Slide the lower wooden wedge in lengthwise direction out of the patient support. If needed use the hammer.
7. Remove also the upper wooden wedge.

Figure 21 - Remove transport provisions



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Remove the screws holding the patient support table on the crate:

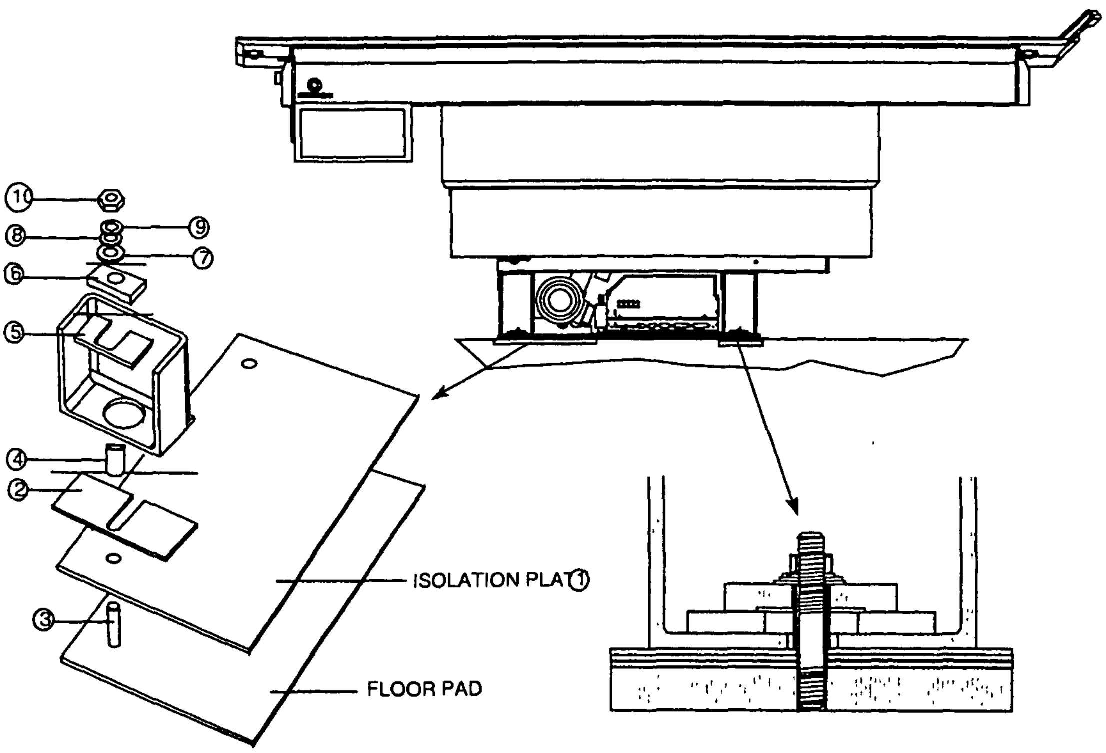
1. Take two 17-mm open-ended spanners, which are needed to remove the screws.
2. Loosen the four screws and nuts, which are used to fixate the patient support table to the crate. Remove the four screws and nuts.
3. Lift the patient support table from the crate and store it in the magnet room, where can be installed when magnet installation is done.
4. Remove the crate and other packing material out of the magnet room. Instruct the movers to take this packing material away from the site.

12.2. MOUNT THE INTERA PATIENT SUPPORT

For the mountıng procedure of the patıent support use drawıng Z3-5.1.2 together wıth the next procedure.

1. Posıtıon the large ısolatıon plate on the pad closest to the magnet.

Figure 22 - Posıtıonıng the Intera patient support



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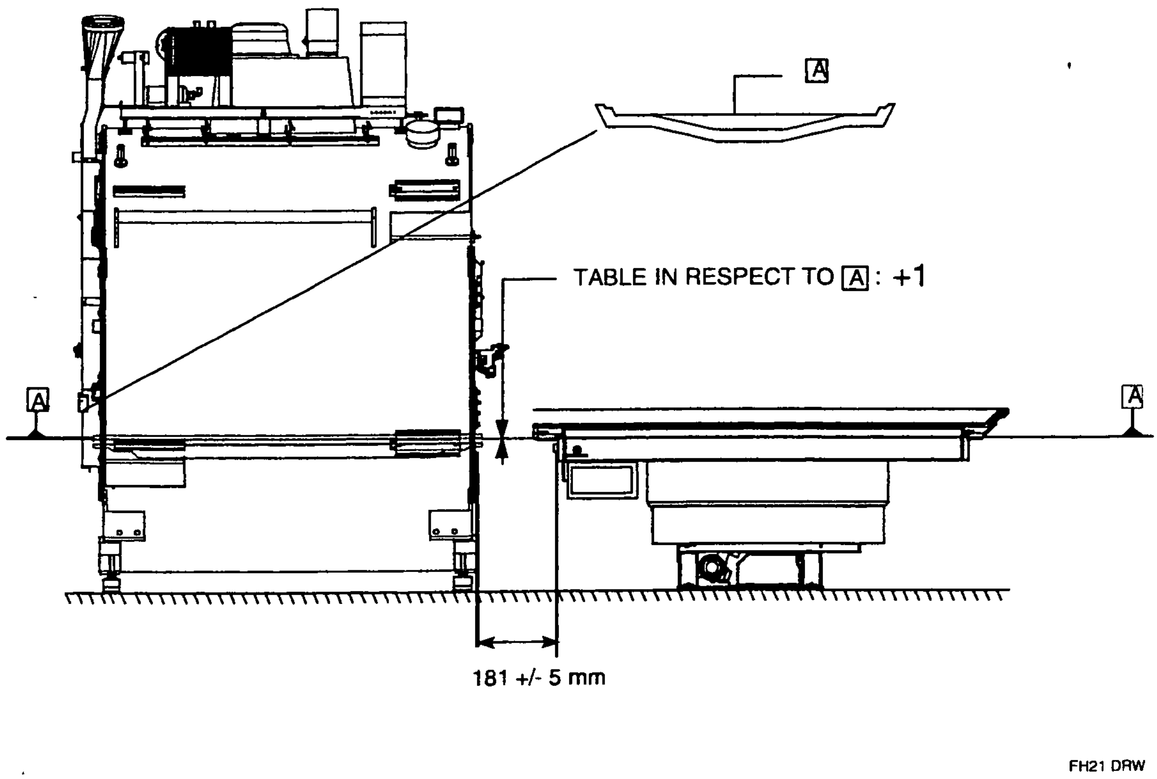
2. Position the patıent support on top of the floor pads, and the large ısolation plate (1).
3. Insert one ısolation spacer (2) between front patıent support frame and floor pad.
4. Mount the four threaded rods (3).
5. Mount the ınsulatıon tubes (4) over the threaded rods (3).
6. Mount the fıxatıon plate (5) (10 mm thıck).
7. Mount fıxatıon block (6). The hole in the fixatıon block ıs not ın the center. Thıs ıs done to enable mounting of this fıxation block wıthın the "Ü" of the u-bar legs. The lower covers wıll be mounted tıghtly against these vertıcal u-bars. Therefor the patıent support table mounting materıal must be ınstalled wıthın the u-bar.
8. Mount ısolatıon washer (7), washer (8), lock washer (9) and nut (10). Do not tıghten the nuts yet.

The "travel-to-scanplane" is a fixed value, therefor the patient support table must be positioned on a given distance from the magnet:

Check/adjust the distance between the patient table and magnet gradient carrier to 181 mm +/- 5 mm This distance is mainly determined during the magnet installation. When the template was used and the magnet has been positioned carefully the distance should be correct. Depending on the accuracy of the patient pads (screw holes) as done by the RF cage supplier, some adjustment is possible by moving the patient support

Figure 23 - Position patient table



12.3. LEVEL THE INTERA PATIENT SUPPORT

12.3.1. INTRODUCTION

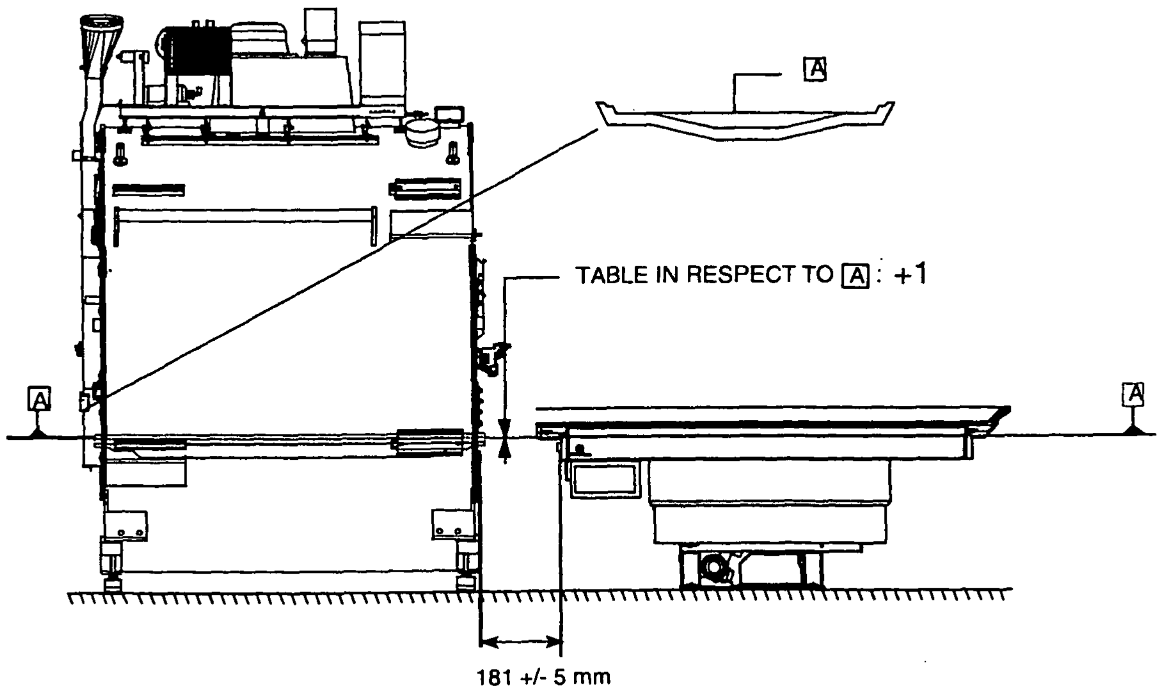
To level the patient table, the bridge section must be installed. Do this as follows:

- Position the "bridge section" with the "finger protection switch plate" towards the patient support.
- Position the four interface points of the "bridge section" above the four holes in the magnet covers.
- Lower the "bridge section " and push it down.

The patient support Intera is delivered in its upper operating position. It is not needed to move the patient support electrically upwards before the height adjustment. Both adjustments can be made without patient support cables connected.

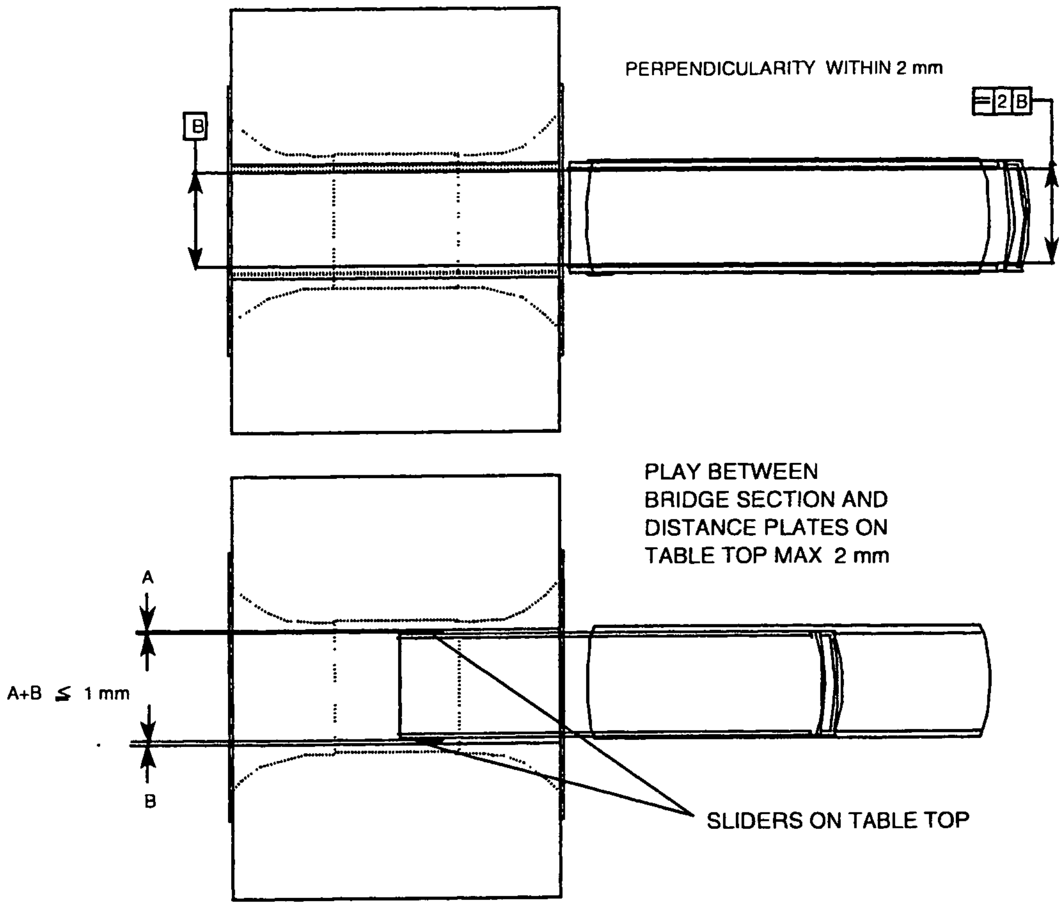
Plane "A" is the running surface of the inner bed. The running surface of the patient support must be 1 mm above the running surface "A" of the inner bed when there is no load on the table.

Figure 24 - Position patient table



Perpendicular alignment between patient support and bridge section must be less than 2 mm.
See adjustment procedure on the next pages

Figure 25- Perpendicular table adjustment



To achieve both the height and the perpendicular alignment perform the following actions:

12.4. ADJUSTMENT OF THE TABLE HEIGHT

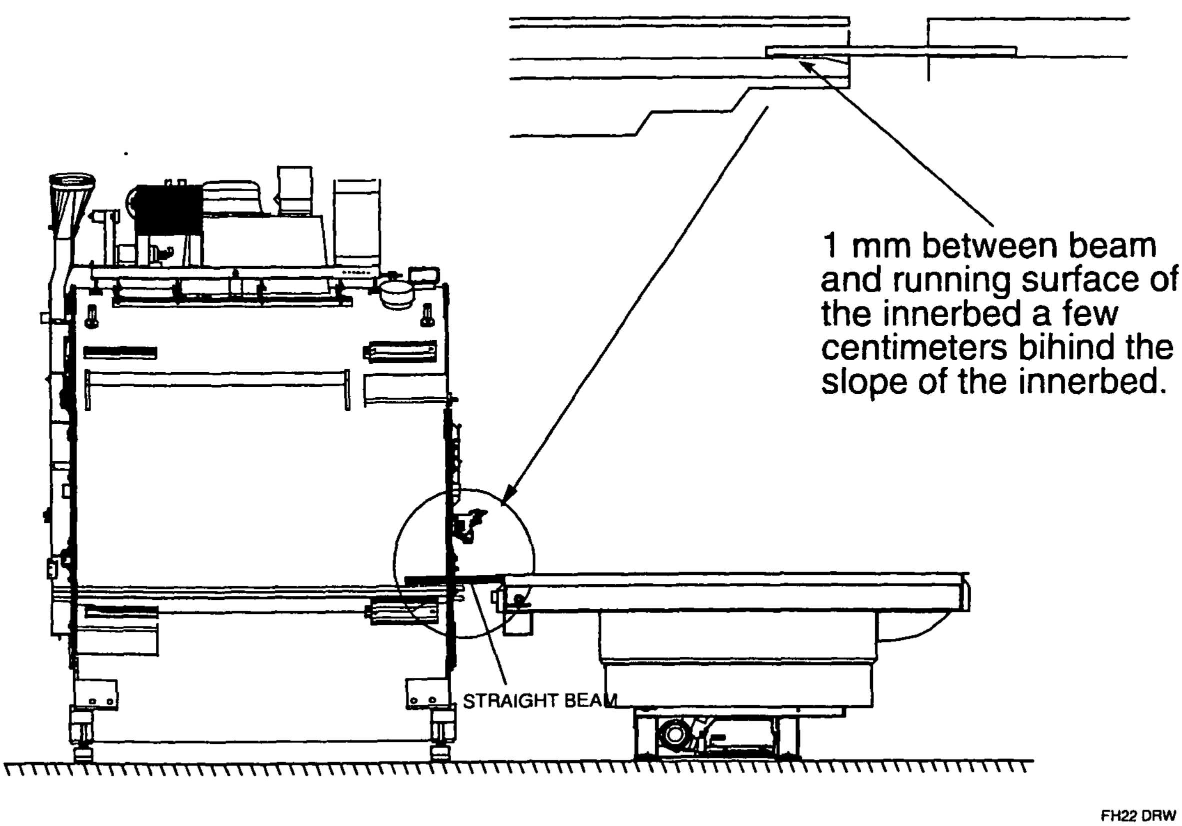
1. Leveling the patient table:

- Level the patient table in horizontally both longways direction and perpendicular on the longways direction using the spirit level. If needed add a spacer to level the patient table

2. The table height adjustment:

  - Mount the bridge section.
    - Position the "bridge section" with the "finger protection switch plate" towards the patient support.
    - Position the four interface points of the "bridge section" above the four holes in the magnet covers.
    - Lower the "bridge section " and push it down.
  - Place the two yellow beams (supplied with the patient support) on top of the running surface of the patient support together with a spirit level. Adjust the height of the patient table (using insulation spacers below the patient table frame) so that there is a 1 mm gap (use a 1 mm spacer) between the beams and the running surface of the bridge section measured a few centimeters behind the sloping surface of the bridge section.

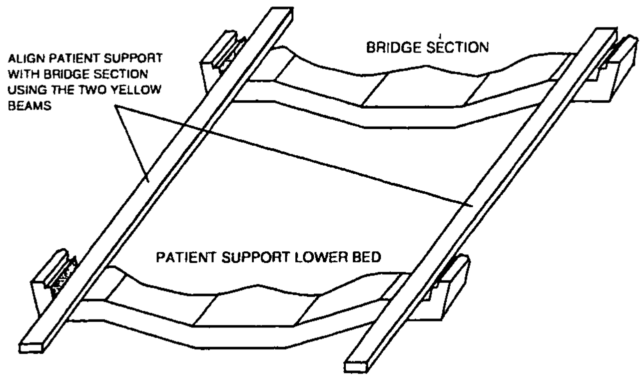
Figure 26 - Adjust table height



12.5. ADJUST THE PATIENT TABLE PERPENDICULARITY

1. The surfaces of the bridge section and patient support lower bed must be in one line (see next figure). Check (by using the two yellow beams) that these surfaces are in one line.

Figure 27 - Align patient support with bridge section

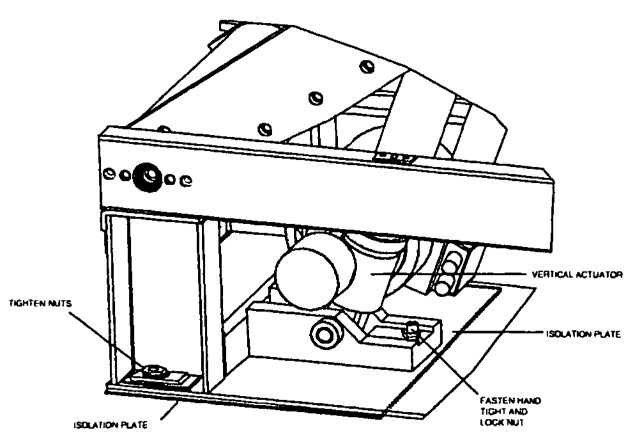


2. Fasten the patient support to the floor by tightening the four nuts.
3. Tighten (hand tight) the threaded rod located near the vertical actuator. In this way, the threaded rod pushes against the isolation plate on the floor so that the patient support frame can withstand downwards forces of the vertical actuator.
4. Lock the nut in order to lock the threaded rod.

CAUTION

Tighten the actuator rod, otherwise the frame can be damaged by placing loads.

Figure 28 - Fasten actuator nut



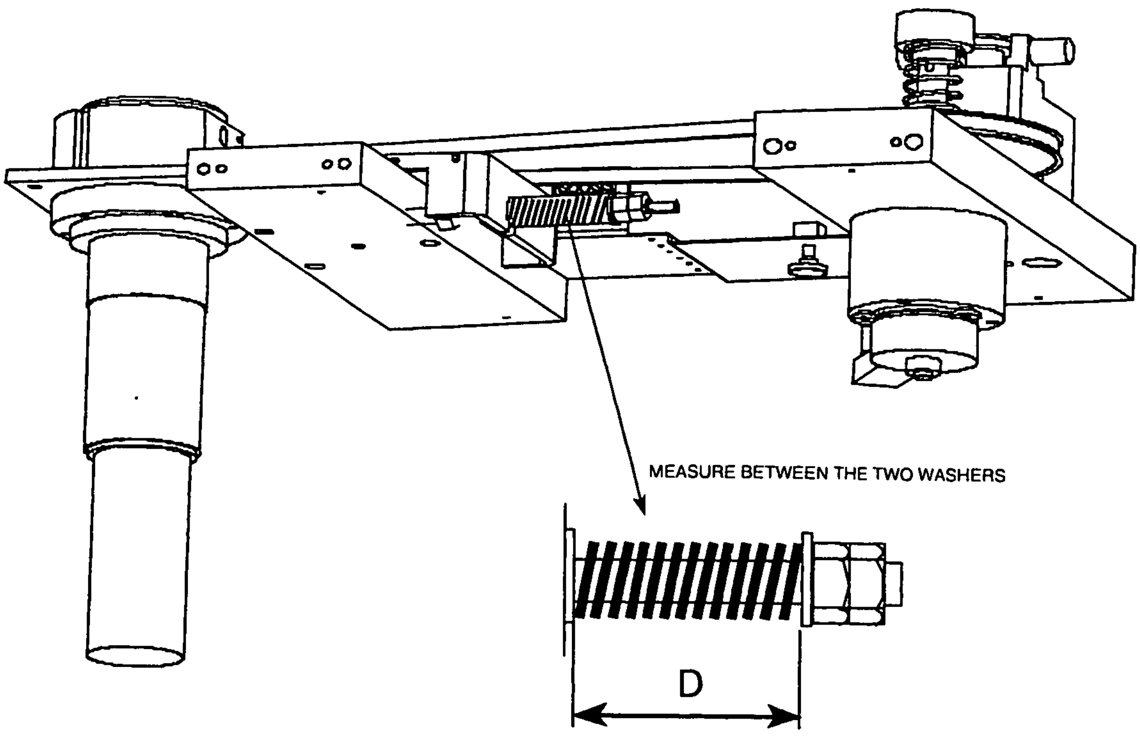
12.6. ADJUST HORIZONTAL TIMING BELT TENSION

The horizontal timing belt must be adjusted by compressing a spring.
Due to the magnet field, a force pulls to the horizontal motor assembly towards the magnet.
For different field strengths, the spring compression is different.

Procedure:

1. Remove the bottom cover underneath the table stretcher.

Figure 29 - Adjust timing belt



2. Adjust the distance D of the spring to:

   42 +/- 1 mm for 1.5 T
   52.5 +/- 1 mm for 1.0 T
   57 +/- 1 mm for 0.5 T (factory adjusted to this value)

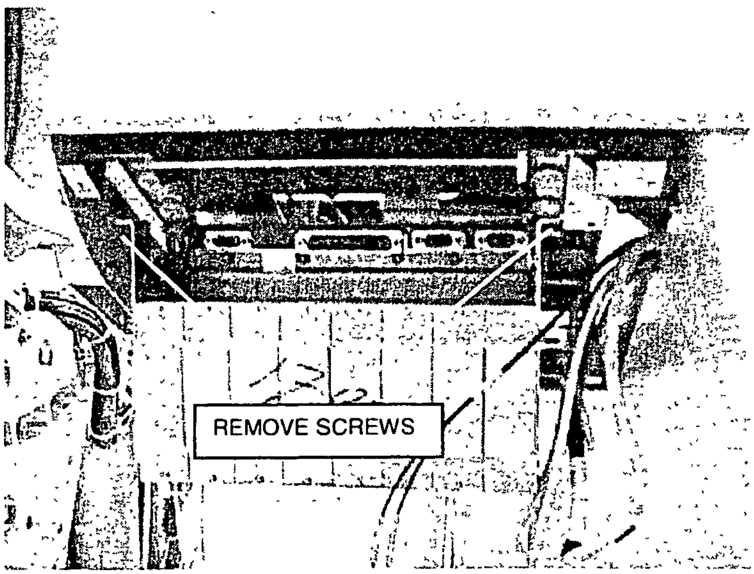
3. Lock the nut with the second nut.
4. Install the cover.

12.7. PHYSIOLOGY RIGHT OR LEFT SIDE

The Physiology unit is default installed at the left side front underneath the patient support.
The physiology unit must always be mounted at the same side as the PICU.
If the PICU is mounted on the right side onto the magnet, than mount the Physiology unit to the right side as follows.

1. Remove the physiology cover from the physiology unit.
2. Remove the lower front cover (magnet side).
3. Unscrew the two screws that hold the physiology unit to the patient support frame The unit will rotate downwards so that you gain access to the cabling.
4. Disconnect all cables.

Figure 30 – Unscrew Physiology unit



5. Dismount the complete physiology unit and mount it at the right side underneath the patient support.
6. Route the cables to the right side and connect them.
7. Mount back the covers.

For mounting of the physiology units, refer to section 19 of this manual.

12.8. MOUNT AND ADJUST THE TABLE TOP

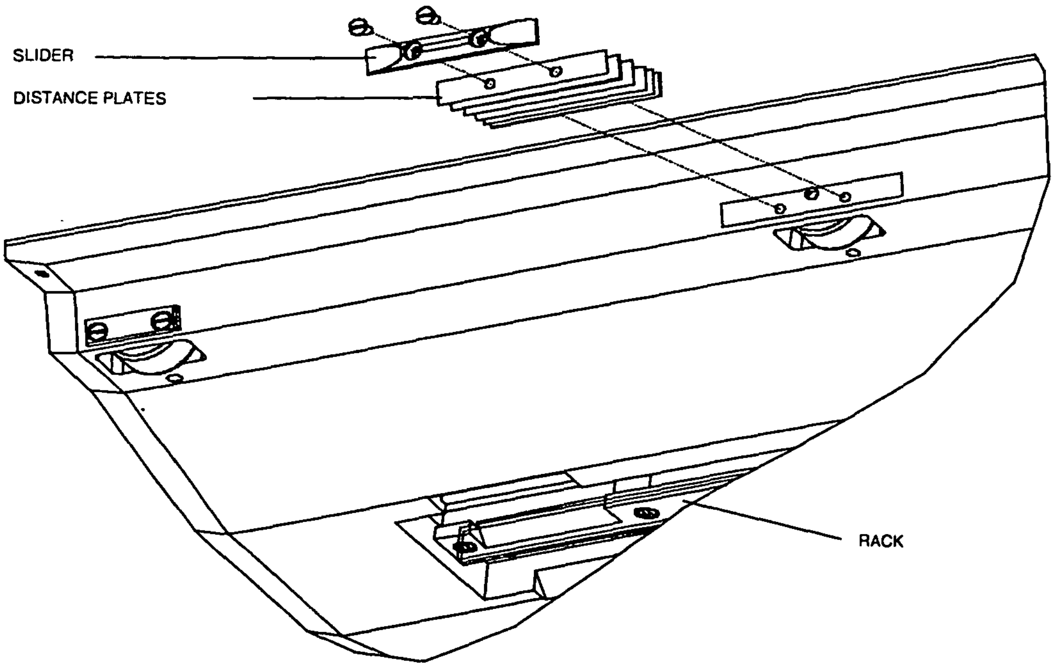
Adjust sideways play:

Maximal 2-mm play is allowed between the table-top and bridge section and between the table-top and table stretcher. Minimize the sideways play as follows:

Check the side-ways play by moving the table-top manually completely in and outwards. Add or remove distance spacers until the table-top sliders are touching the bridge section or/and table stretcher. Than remove one spacer on the right and left side of the table-top.

It may not always be possible to add an equal amount of distance plates on right and left side. This is caused by the table-top position, which is not always symmetrical between the table stretcher.

Figure 31 - Adjust side ways play



Check the alignment without load, and with a load of ± 100-kg, by moving the tabletop in and out by hand. No bumping or scraping must be felt.

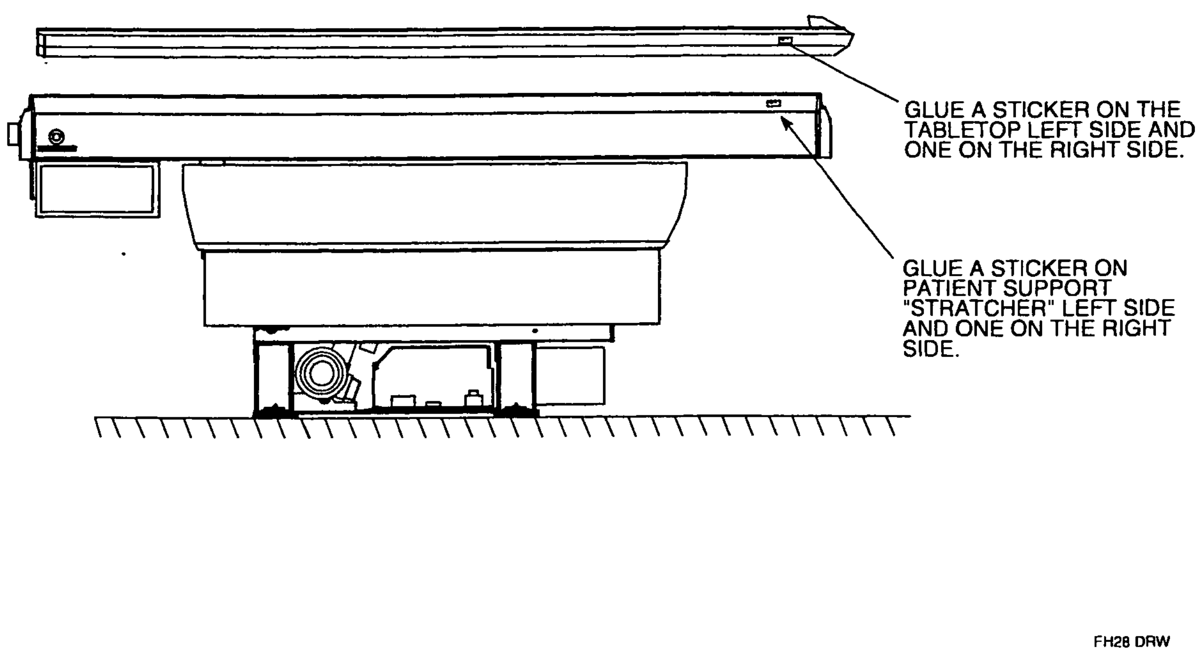
12.9. IDENTIFICATION STICKER

A tabletop Release 7 that is adjusted for a particular system cannot be used on another system.
The yellow tabletop for Release 7 can be used on a trolley.

If a hospital has two or more systems near to each other, and a trolley option is used, an operator must be able to determine which tabletop belongs to which system. Identification stickers are therefor supplied with each tabletop. These stickers must be glued on the patient table and on each tabletop.

Glue one sticker (MR1) on the patient table of system 1. Glue another sticker [MR1] on a tabletop belonging to system 1. If there is more than one table top for system 1 then also glue a sticker [MR1] on those tabletops. Do the same for system 2 (sticker [MR2]) and system 3 (sticker [MR3]) if present.

Figure 32 - Location of identification sticker



12.10. ONE TROLLEY FOR MORE SYSTEMS

If a trolley release <6.2 (4522 131 06814) or a trolley release 6.2 (4522 131 61701) is present in the hospital, than this trolley must be modified. During installation **check if old trolleys are present** and if it is present, be sure that it is modified using the upgrade kit.

WARNING

The table-top for Release 7 may not be used on an old type trolley. ***This creates an unsafe situation*** *that must be avoided. The table-top can roll on the trolley. There is a risk of fingers getting caught between handlebars of trolley and table-top. Be sure to update the old trolleys.*

In the commercial catalog, the customer is asked weather he has an old type trolley. If the answer is YES, a modification kit will be send together with the initial system.

12.11. FINAL CHECK

1. Measure with an Ohmmeter that the patient support frame is electrically separated from the RF enclosure. You can measure between table frame and floor pads. The measured value must be more than 200 KOhm. Write down the measured value into the MRL.
2. Connect all cabling.
3. Mount the lower cover. Make sure to connect the safety grounds to these lower covers.

13. MOUNTING THE EXTERNAL COOLING SYSTEM TO THE TOP AIR DUCTS EXTENSIONS.

Mount the external air hoses (non-Philips) to the air ducts Hoses must be fixed using hose clamping rings (non-Philips)

CAUTION

Take care that the air ducts and hose clamping rings do not vibrate against other objects like the false ceiling. This can cause image artefacts due to spikes.

14. MOUNTING THE PATIENT COOLING AIR DUCT

Mount the (delivered) patient cooling air duct between the patient cooling air duct at the magnet rear and the patient cooling unit inside the technical room.

Note that the air for the inlet of the patient cooling unit is taken out of the examination room. This is done to avoid disturbance of the air conditioning system on the examination room.

Section 7 Mechanical installation (Part 2)

Contents

1. INTRODUCTION

This section describes the mechanical installation of the following units:

Figure 1 – Magnet right side view

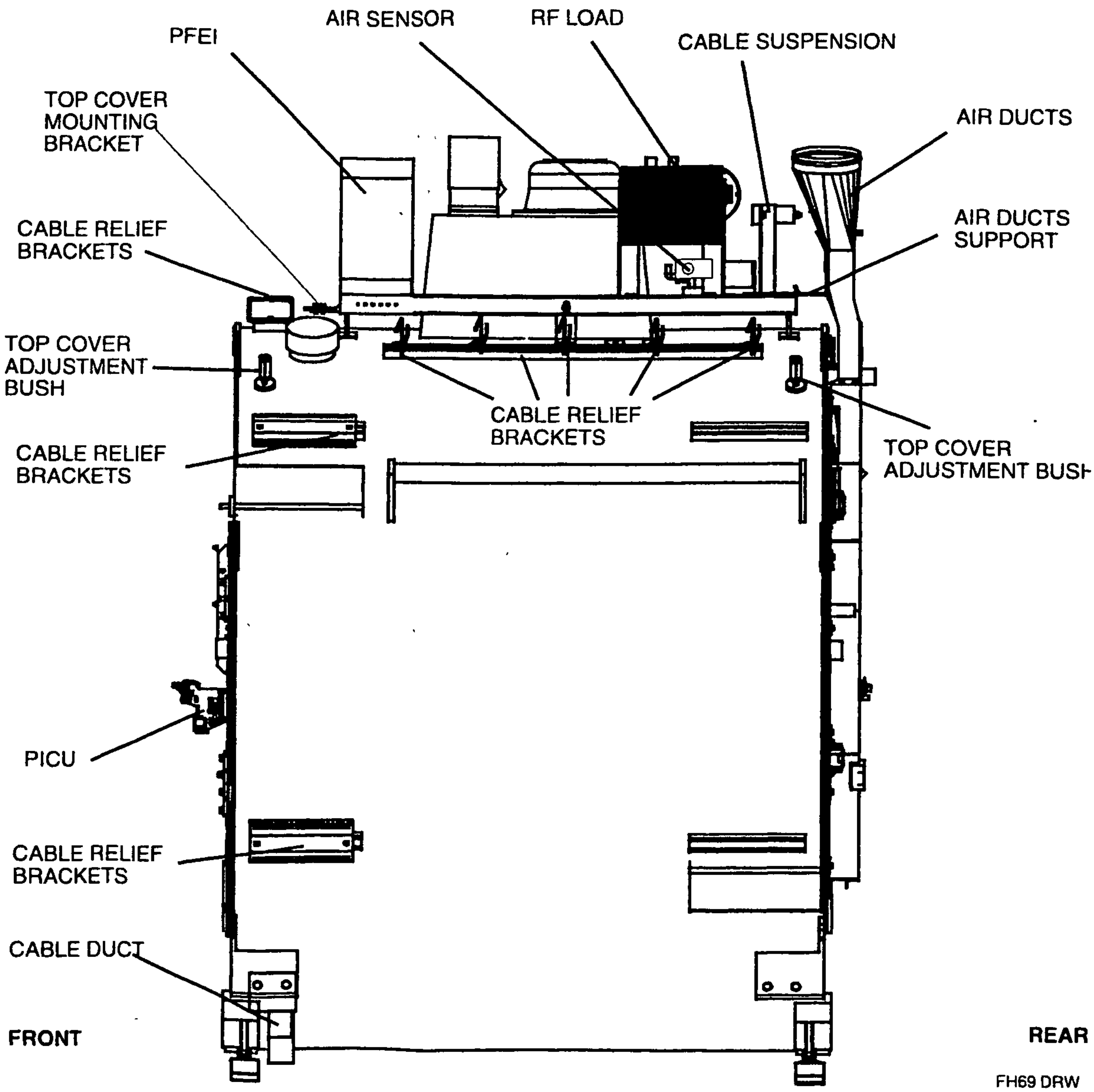


Figure 2 – Magnet rear view

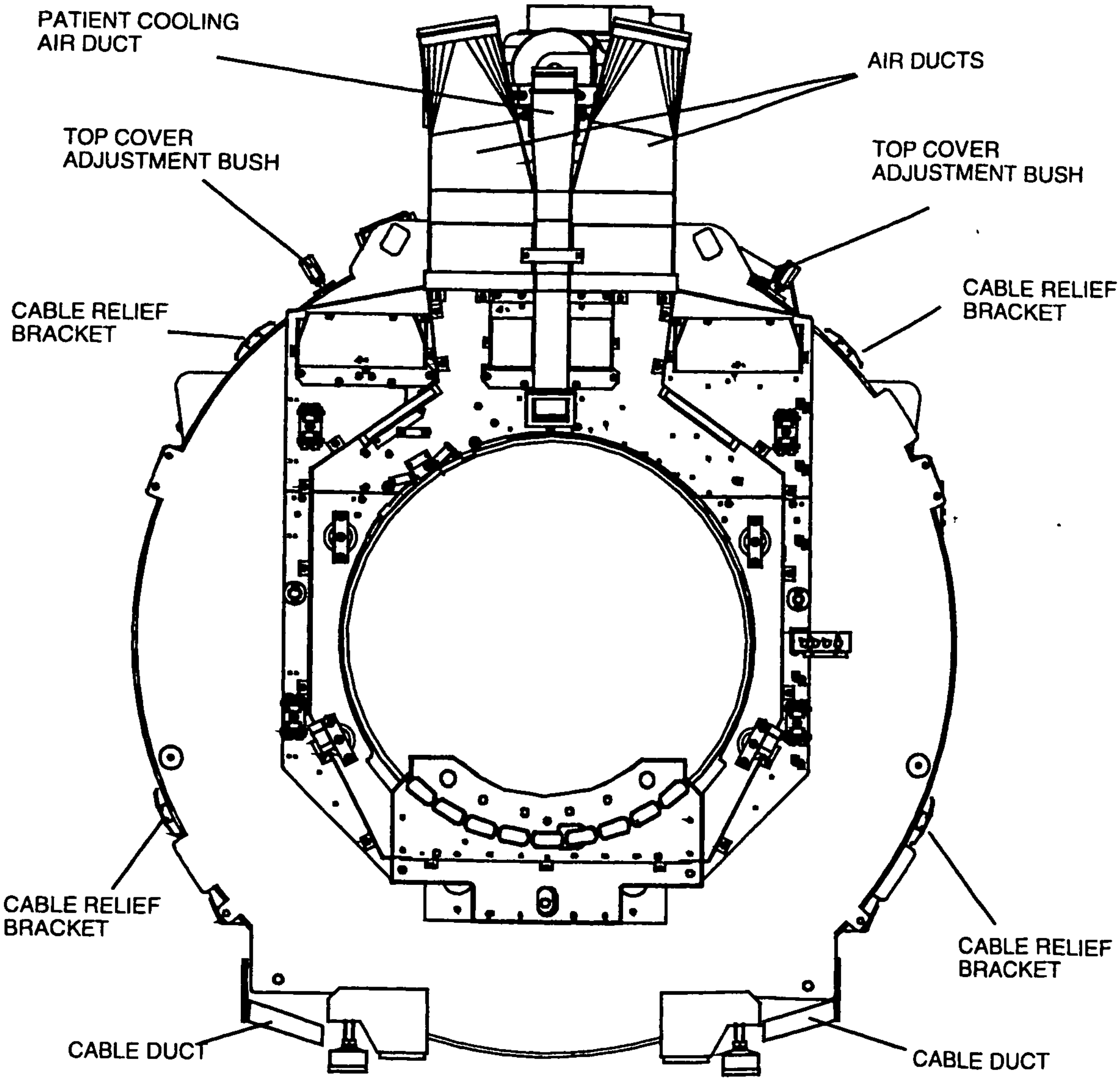


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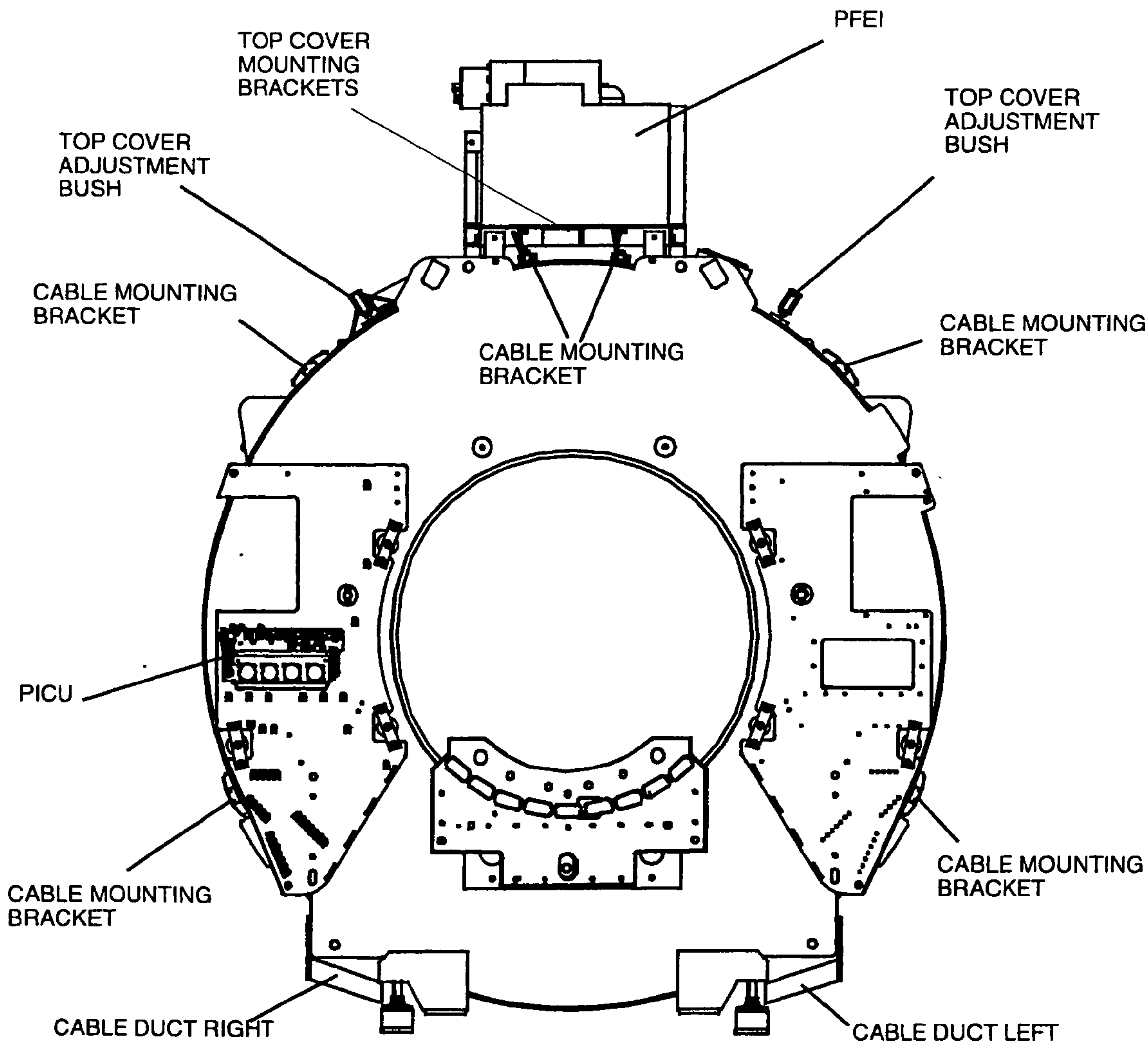
Figure 3 – Magnet Front view



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2. MOUNTING THE PFEI

Mount the PFEI on top of the magnet top-frame at the front side of the magnet.
The service key stickers on the PFEI must point towards the patient end.

Figure 4 – Position the PFEI

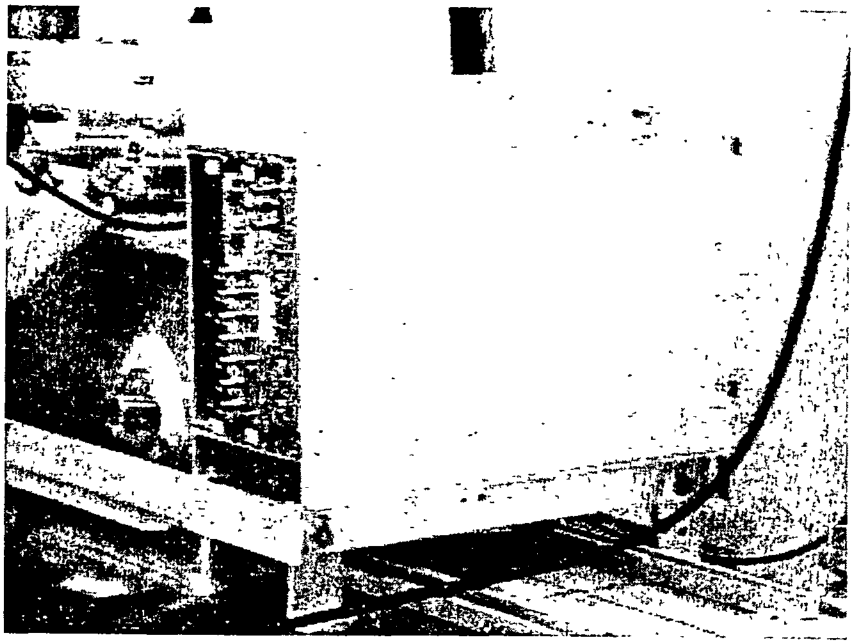
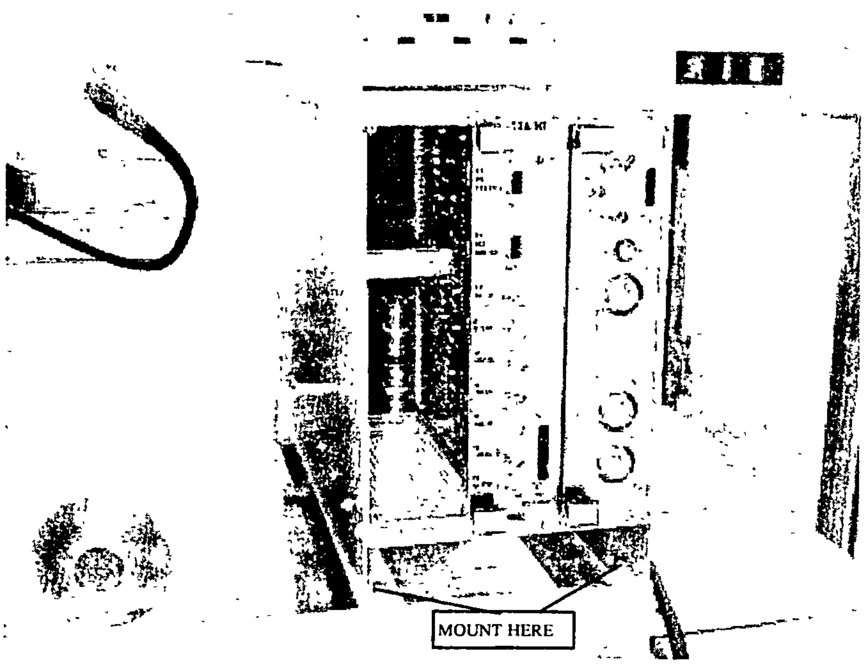
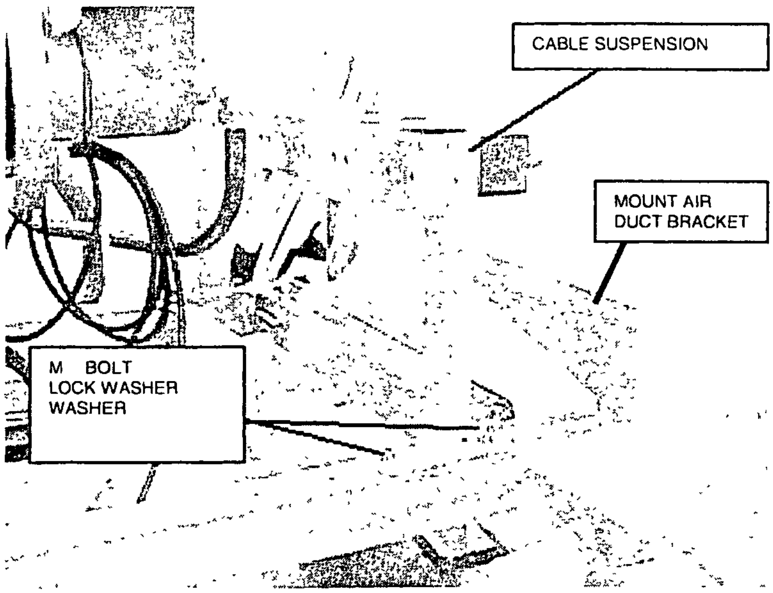


Figure 5 – Mount the PFEI



3. MOUNTING THE CABLE SUSPENSION

Mount the cable suspension on top of the magnet top-frame at the rear side of the magnet.

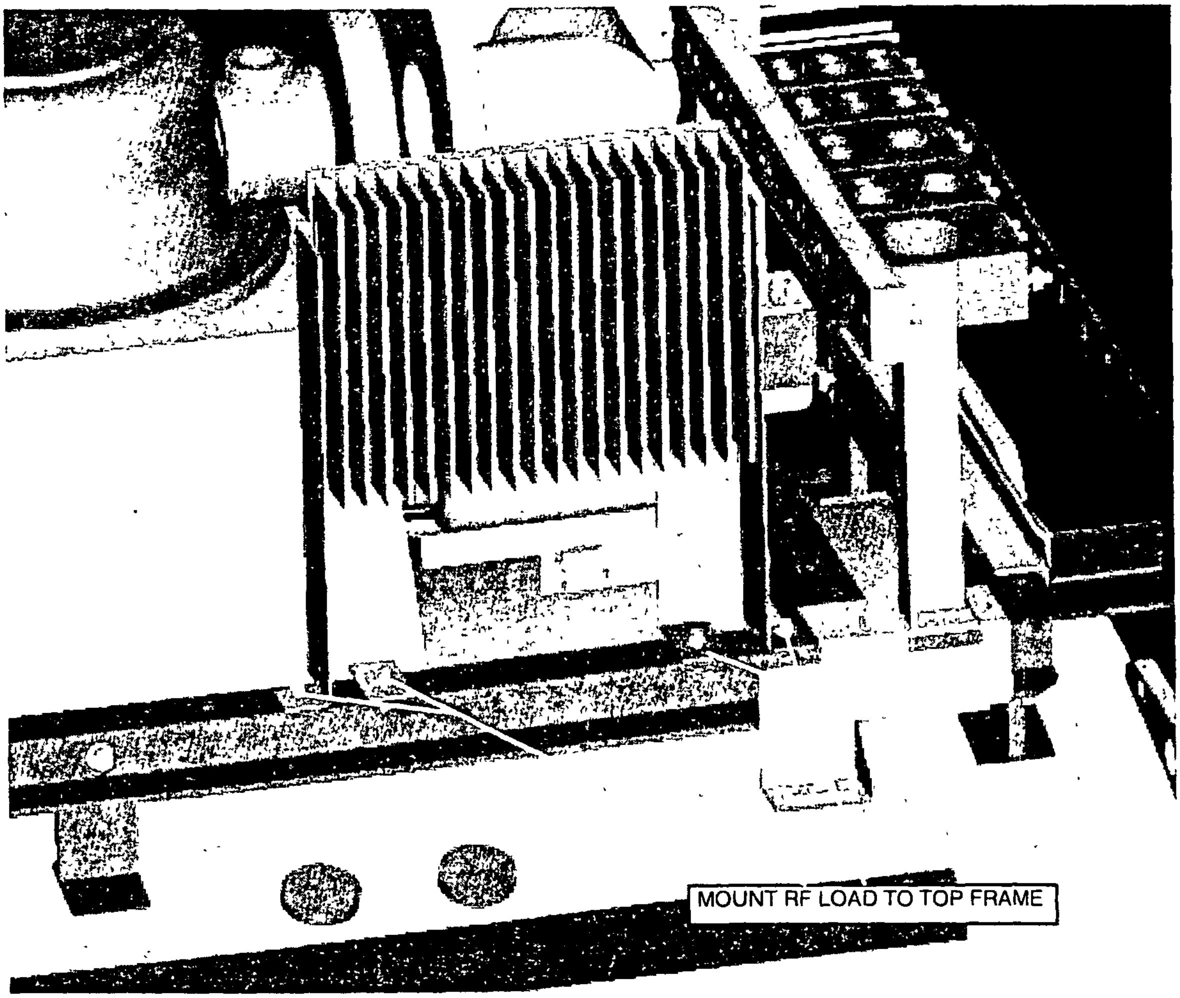


4. MOUNT THE RF LOAD ASSY

Mount the RF load assy on top of the magnet top-frame at the magnet right/rear side.

Mount the RF load with the RF load connector towards the center of the magnet otherwise the top covers will not fit.

Figure 6 – Mounting the RF load



5. MOUNTING THE CABLE RELEAF BRACKETS

Four cable relief brackets must be mounted to the magnet. Two brackets at the left side and two at the right side.

Figure 7 – Cable bracket (left bottom side)

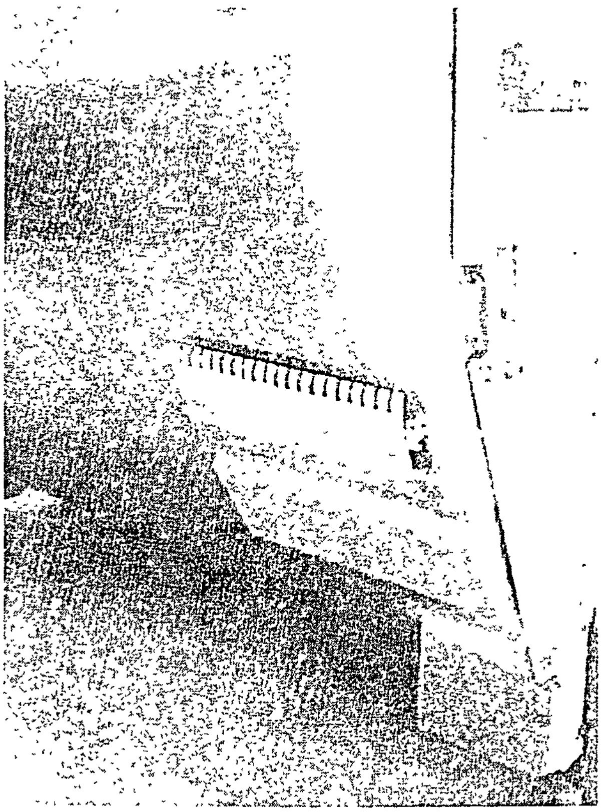
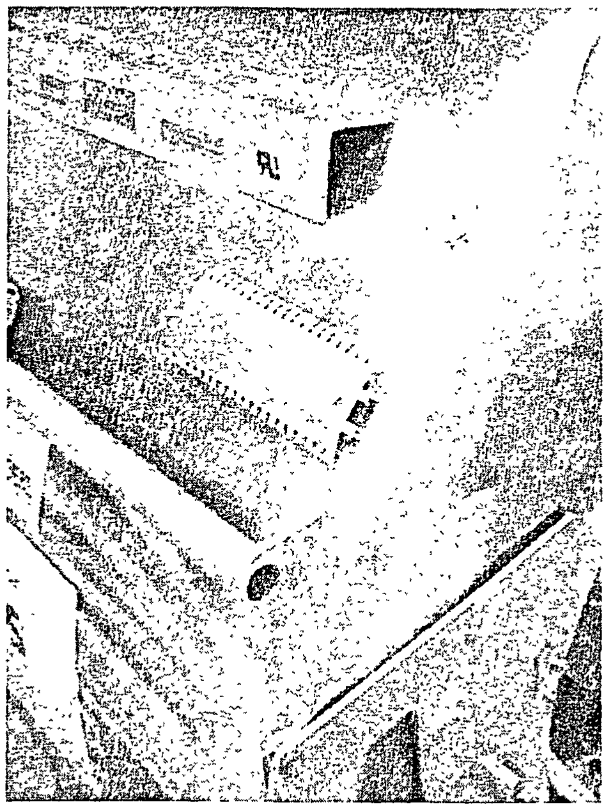
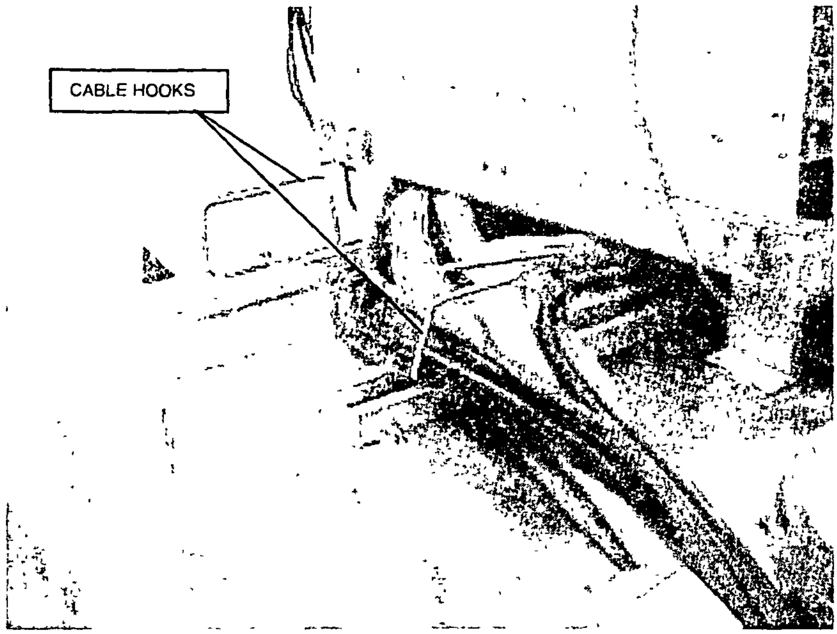


Figure 8 – Cable bracket (left top side)



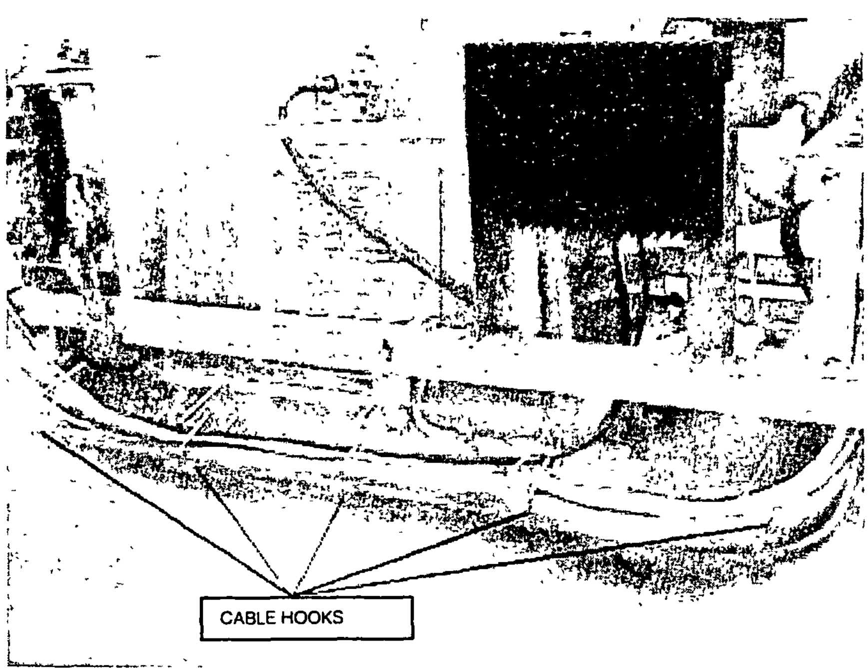
Mount two cable hooks at the top front side of the magnet (before the PFEI).

Figure 9 – Mount cable hooks



Mount four cable hooks to the magnet at the right topside.

Figure 10 – Mount cable hooks



6. MOUNTING THE TOPCOVER BRACKETS

1. Mount the top-cover bracket at the front of the magnet against the magnet top frame.

Figure 11 – bracket at magnet front

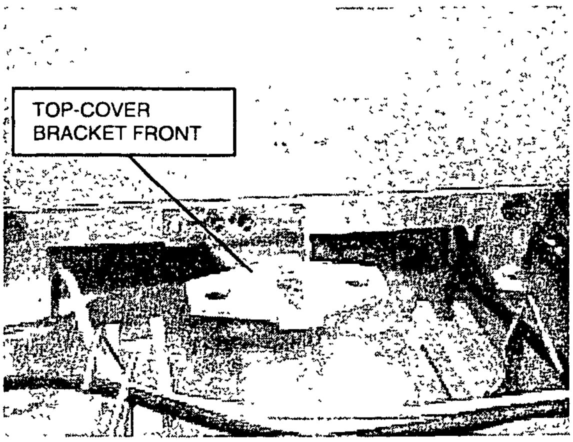
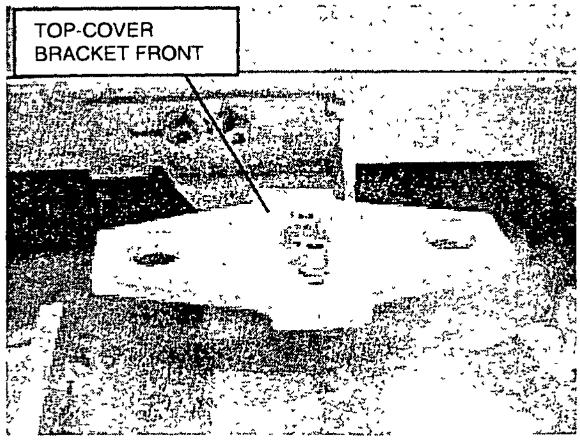


Figure 12 - bracket at magnet front



2. Mount two top cover brackets together with the cable suspension

Figure 13 – Top cover brackets

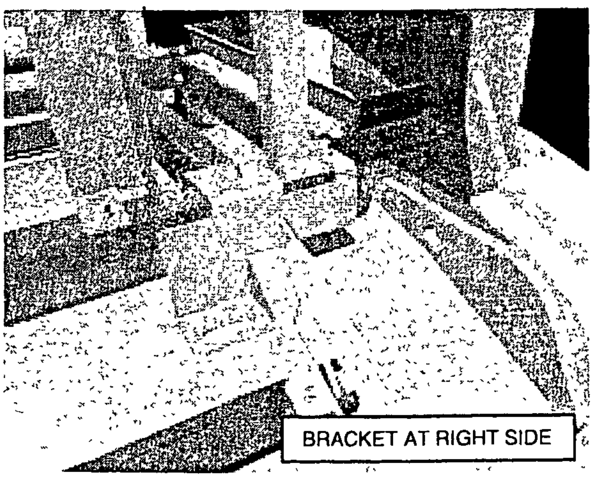
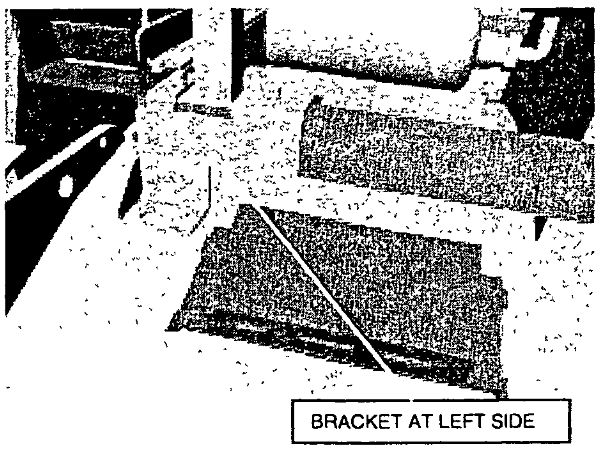
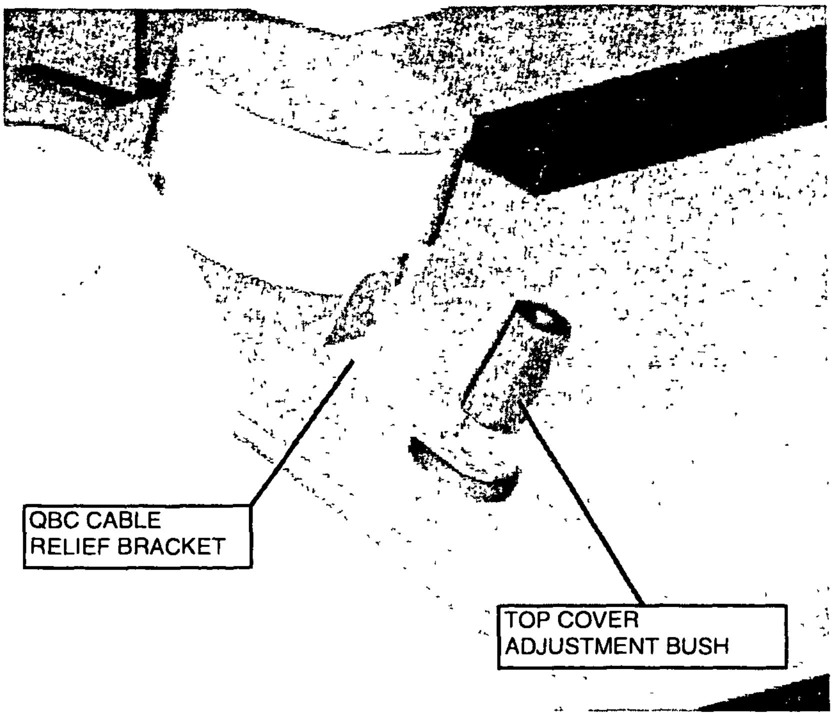


Figure 14 – Top cover brackets



3. Mount four top cover adjustment bushes on top of the magnet.
   At the front right side of the magnet, mount the "QBC cable relief bracket". Mount this bracket together with the "top cover adjustment bush".

Figure 15 - Top cover adjustment bush



7. MOUNTING THE CABLE DUCTS

At the left and right side of the magnet, mount a cable duct
The cables towards the patient support are rooted trough these cable ducts

Figure 16 – Location of cable duct

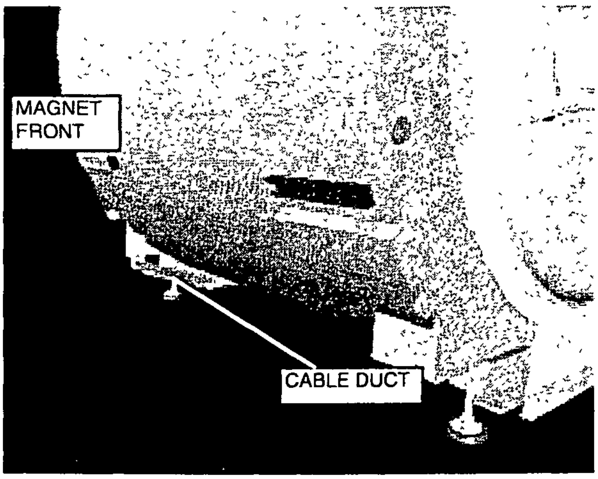
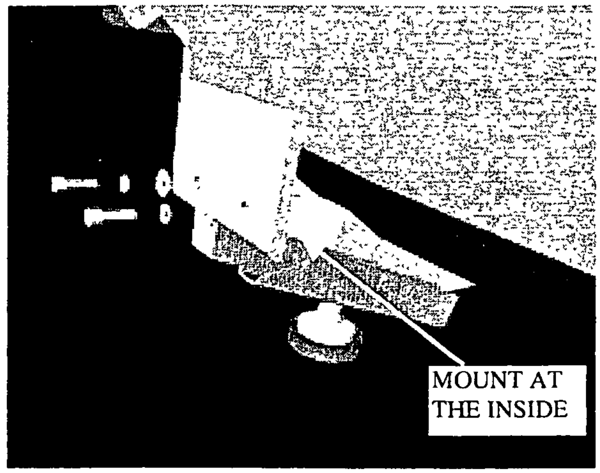


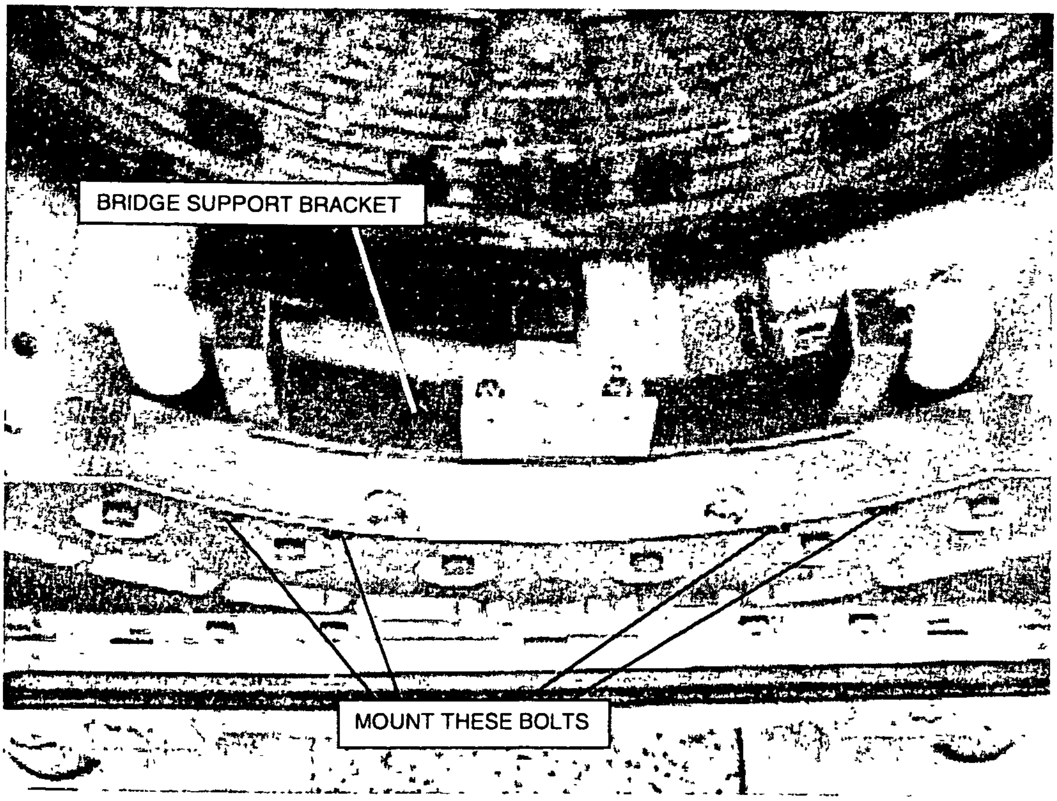
Figure 17 – Mounting the cable duct



8. MOUNTING THE BRIDGE SUPPORT BRACKETS

Mount the front bridge support brackets (bracket with wire for the switch plate) against the front gradient carrier. Mount the other bridge support bracket against the rear gradient carrier.

Figure 18 – Mounting the bridge support bracket



9. MOUNTING THE GRADIENT STRIPS

Skip this paragraph if you have a factory assembled magnet system

9.1. GENERAL

The gradient strips must be mounted on the rear ring frame. For installation details, refer to drawing Z6-4.8 and Z6-4.9 of the drawing manual.

NOTE

Special attention must be paid to ensure that the surfaces are clean and free from dirt and/or resin. Also the fibre glass insulation must not get caught under the cable.
If the insulation needs trimming use a small sharp scalpel. You must wear safety goggles in case the blade breaks.

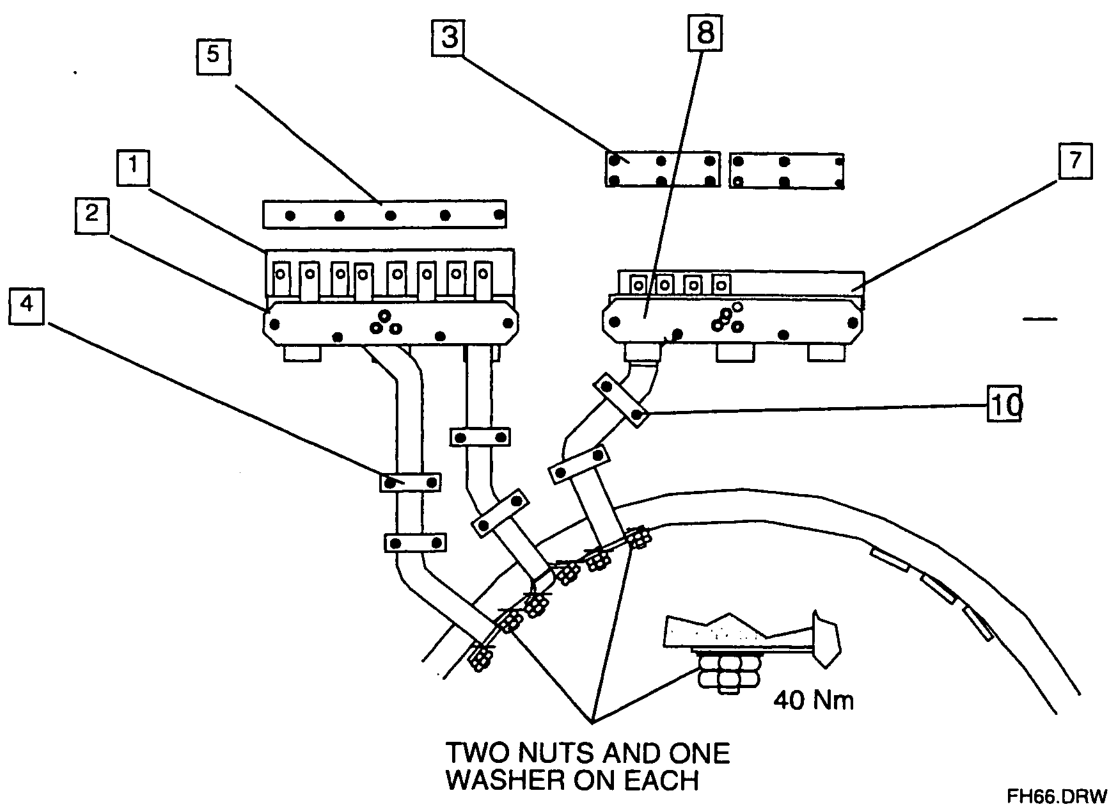
Mount the strips to the gradient coil as follows: Torque the first nut with **40 Nm** using a torque wrench. Apply also 40 Nm to the second nut without touching the first nut.

9.2. PROCEDURE FOR POWER GRADIENTS

Use drawing Z6-4.8

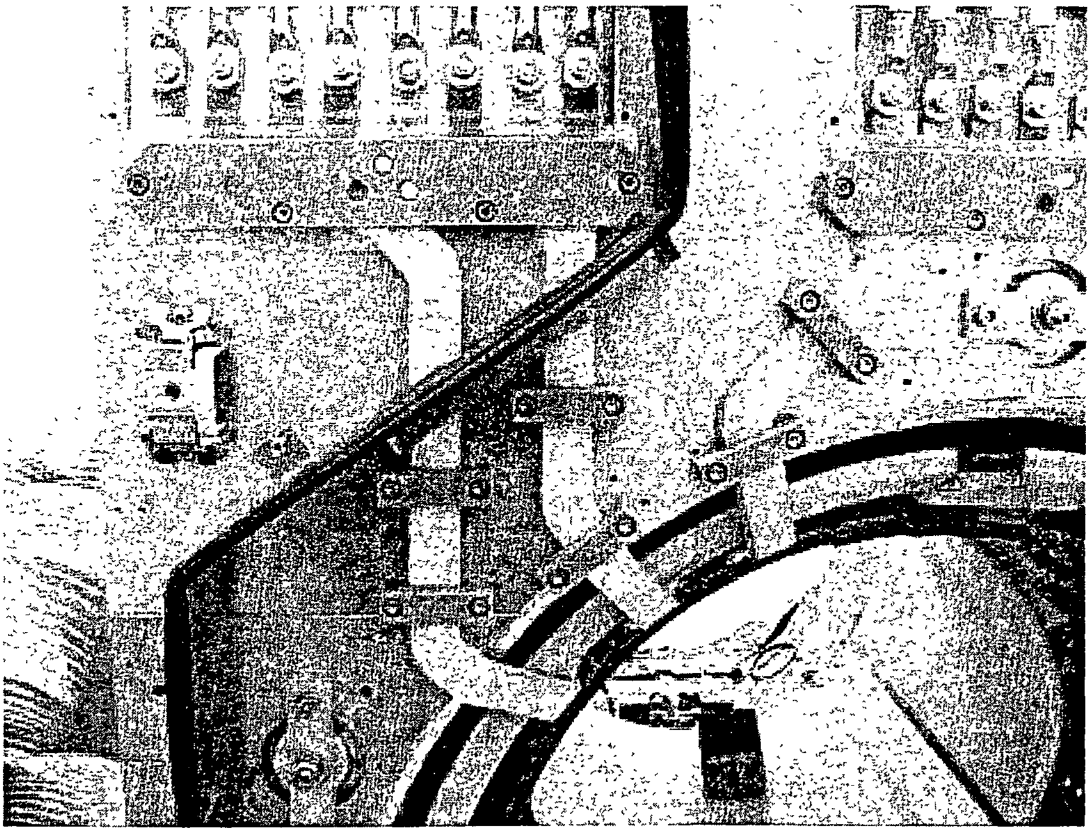
1. Remove the nuts from the gradient coil.
2. Remove the left upper partition top (aluminum plate) from the ring-frame.
3. Attach the insulation plate [1] to the ring-frame. Use some tape to keep it in position.
4. Mount the most left strip assy [2] as follows:
   - First mount the assy to the gradient coil (one washer and two nuts). Do not tighten the nuts.
   - Fix the assy to the ring-frame (blocks [2]). Do not tighten the bolts.
   - Remove the aluminum plate [3] (Z6-8) from the assy. This aluminum plate is no longer needed.
   - Mount the clamping blocks [4]. Do not tighten the bolts.
5. Mount clamping blocks [6].
6. Attach insulation plate [7] to the ring-frame. Use some tape to keep it in position.
7. Mount the middle strip assy [8] as follows
   - First mount the assy to the gradient coil (one washer and two nuts). Do not tighten the nuts.
   - Fix the assy to the ring-frame (blocks [8]). Do not tighten the bolts.
   - Remove the aluminum plate [9] (Z6-4.8) from the assy. This aluminum plate is no longer needed.
   - Mount the clamping blocks [10]. Do not tighten the bolts.
8. Mount plastic safety cap [13] (Z6-4.8)
9. Mount the left upper partition (aluminum plate) to the ring-frame. Note that two foam blocks are used to close the additional holes in the partition top.
10. Tighten all bolts. Start at the top [2] and [8], than the clamping blocks [4] and [10] and at last the gradient coil connections. Tighten the gradient coil nuts each with **a torque of 40 Nm** using a torque wrench.

Figure 19 - Gradient strips for the power gradients



FH66.DRW

Figure 20 – Gradient strips for the Power Gradients



11 Measure with an ohmmeter if no short circuit is present between the strips. Measure between the following points on the gradient coil:

XB+ - YB+
YB+ - ZB+
ZB+ - XB+

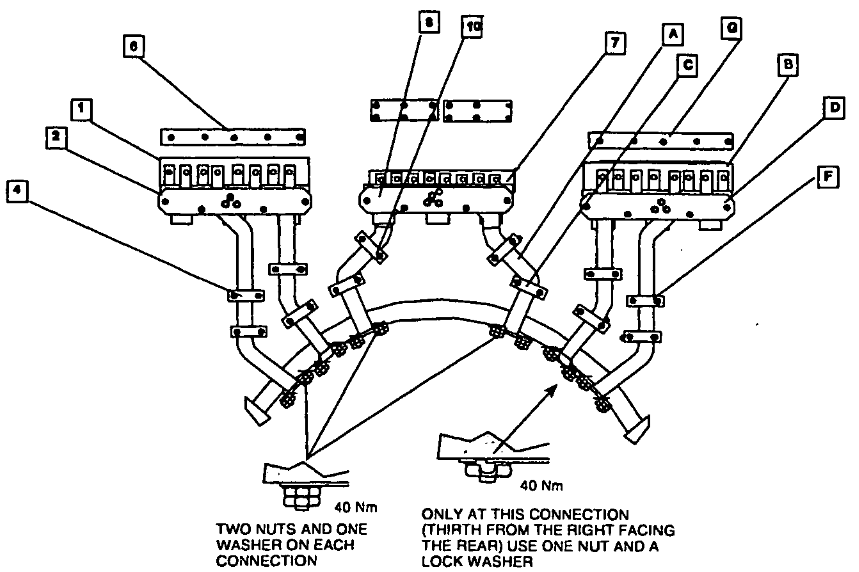
You should measure >10 Ohm.
Measure also between the each strip and the ring-frame. Also here you should measure >10 Ohm.

9.3. PROCEDURE FOR THE MASTER GRADIENTS

Use drawing Z6-4.8 and Z6-4.9

1. Remove the nuts from the gradient coil.
2. Remove the upper right and left partition top (aluminum plate) from the ring-frame.
3. Attach the insulation plate [1] (Z6-4.8) to the ring-frame. Use some tape to keep it in position.
4. Mount the most left strip assy [2] as follows:
   - First mount the assy to the gradient coil (one washer and two nuts). Do not tighten the nuts.
   - Fix the assy to the ring-frame (blocks [2]). Do not tighten the bolts.
   - Remove the aluminum plate [3] from the assy. This aluminum plate is no longer needed.
   - Mount the clamping blocks [4]. Do not tighten the bolts.
5. Mount clamping blocks [6].
6. Attach insulation plate [7] to the ring-frame. Use some tape to keep it in position.
7. Mount the two strips [A] (Z6-4.9) into the middle strip assy [8] (Z6-4.8 and Z6-4.9). Take care that the isolation plates are in the correct position.
8. Mount the middle strip assy [8] as follows:
   - First mount the assy to the gradient coil. (One washer and two nuts). Do not tighten the nuts.
   - Fix the assy to the ring-frame (blocks [8]). Do not tighten the bolts.
   - Remove the aluminum plate [9] (Z6-4.8] from the assy. This aluminum plate is no longer needed.
   - Mount the clamping blocks [10] (Z6-4.8) and [C] (Z6-4.9). Do not tighten the bolts.
9. Attach insulation plate [B] to the ring-frame. Use some tape to keep it in position.
10. Mount the right strip assy [D] as follows:
    - First mount the assy to the gradient coil (one washer and two nuts except for the 3rd connection from the right. Use here a lock washer and one nut). Do not tighten the bolts.
    - Fix the assy to the ring-frame (blocks [D]). Do not tighten the bolts.
    - Remove the aluminum plate [E] (Z6-4.9) from the assy. This aluminium plate is no longer needed.
    - Mount the clamping blocks [F]. Do not tighten the bolts.
    - Mount clamping blocks [H] with insulation strip [G].
11. Tighten all bolts. Start at the top [2], [8] and [D]. Then fix the clamping blocks [4], [10] [C] and [F]. At last tighten the gradient coil nuts. **Apply 40 Nm** to each nut using a torque wrench.
12. Bend the lock washer using a hammer and a screwdriver. Bend one end against the nut and the other end over the edge of the gradient strip (see part 8 on drawing Z6-4.9).

Figure 21 – Gradient strips for the Master gradients



FH65.DRW

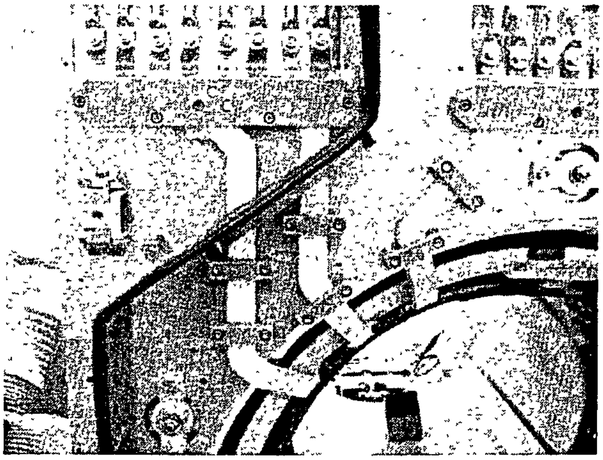


Figure 22 – Gradient strips (Master gradients)

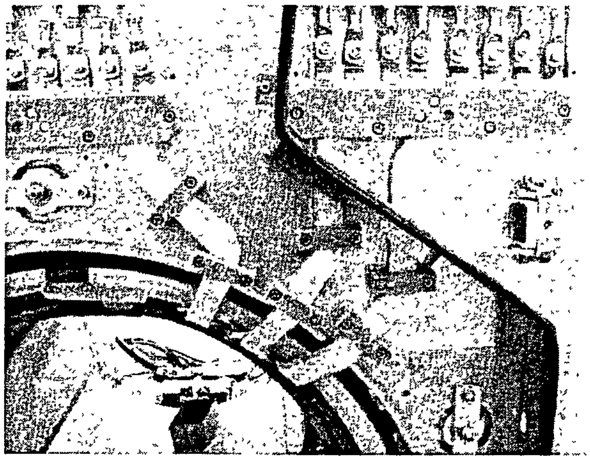


Figure 23 – Additional strips (Master gradients)

13. Measure with an ohmmeter if no short circuit is present between the strips. Measure between the following points on the gradient coil:

    YA+ - XB-
    YA+ - ZB-
    XB- - ZB-
    YA+ - XA+
    YA+ - ZA+
    XA+ - ZA+
    YB+ - XB+
    YB+ - ZB+
    XB+ - ZB+

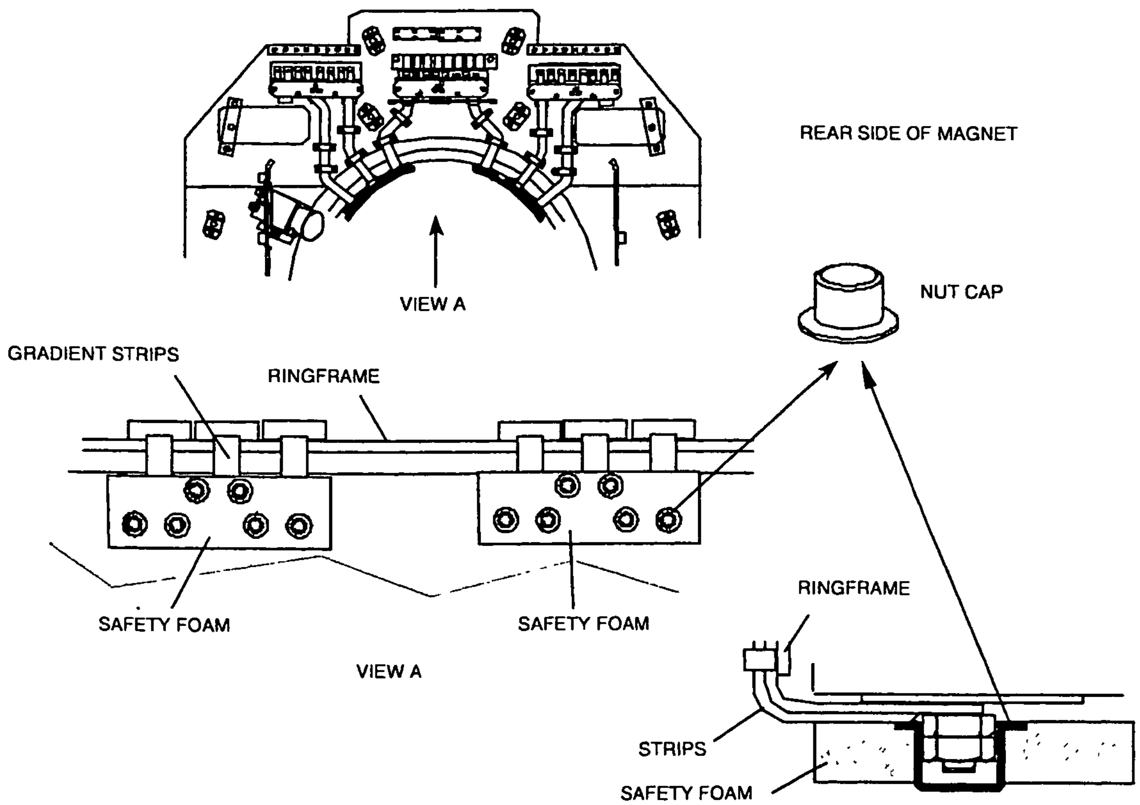
    You should measure >10 Ohm.
    Measure also between the each strip and the ring-frame. Also here you should measure >10 Ohm.

9.4. MOUNT THE SAFETY PROTECTION FOAM

Skip this paragraph if you have a factory assembled magnet system

For safety reasons, foam must be glued on top of the gradient strips. This must prevent an engineer from accident touching the gradient coil connections while the gradient amplifier is switched ON.

Figure 24 – Glue safety protection foam



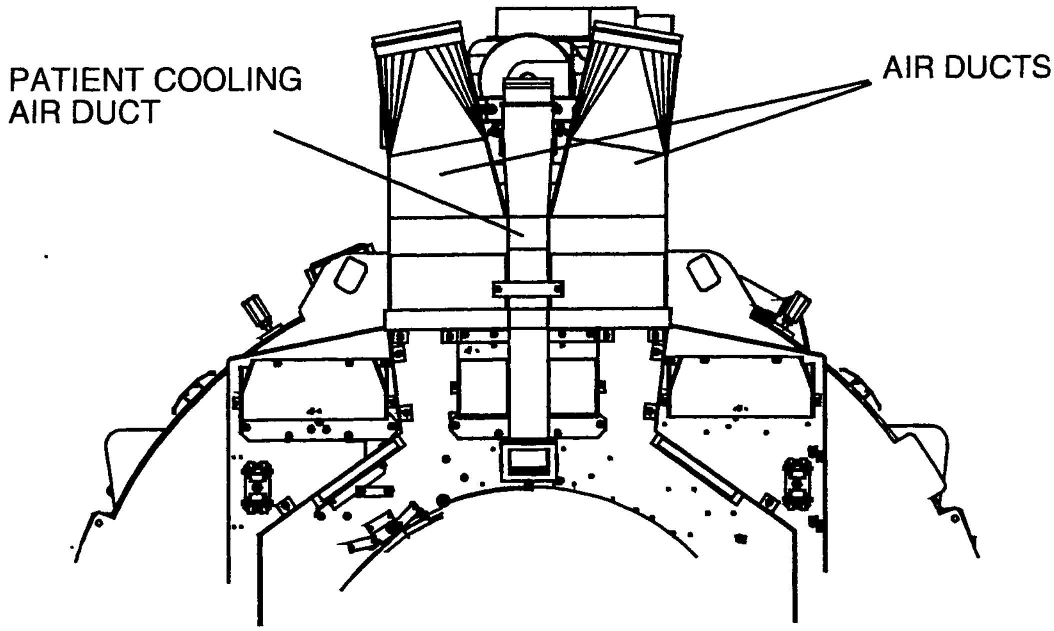
Remove the plastic layer, mount the nut caps into the foam and glue the foam to the gradient strips.
The nut caps are positioned over the nuts.

10. MOUNTING THE AIR DUCTS

Skip this paragraph if you have a factory assembled magnet system

The gradient cables must be mounted before mounting the air ducts. Mount first the left and right air ducts and than the patient cooling air duct.

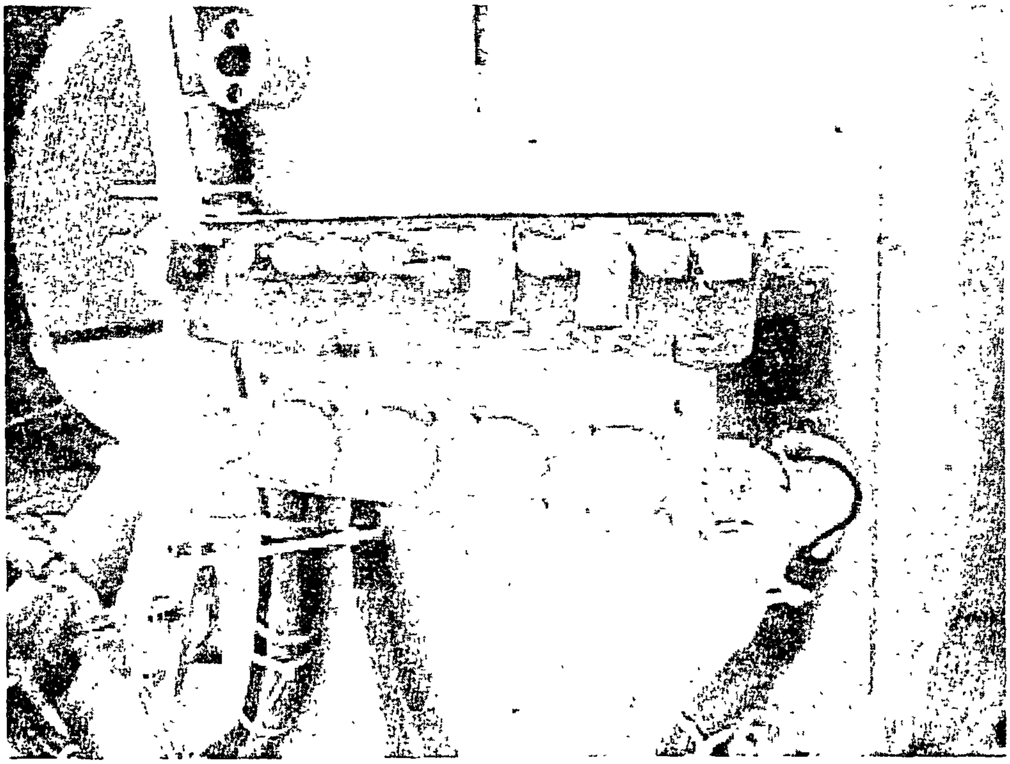
Figure 25 – Air ducts



11. MOUNTING THE PATIENT INTERFACE CONTROL UNIT (PICU)

The PICU can be installed at the left or right side of the front ring-frame. This position is site dependent, so ask the hospital staff where they want to have it mounted. Note that the PICU must be at the same side as the Physiology unit in the patient support Also note that a trolley must be docked at to the patient table at the opposite side of the PICU and Physiology unit.

Figure 26 – Mount the PICU



The rear PICU is already mounted onto the rear ring-frame. For factory pre-assembled magnet systems the rear PICU is attached to the rear ring-frame with cable binders. In that case cut the cable binders and mount the rear PICU to the rear ring-frame.

12. INSTALL THE INTERA PATIENT SUPPORT

NOTE

Before shimming the magnet, the patient support must be positioned in front of the magnet in the upmost position.

12.1. REMOVE TRANSPORT PROVISIONS

To prevent damage to the patient support, wooden wedges are mounted at the right and left side into the patient support.

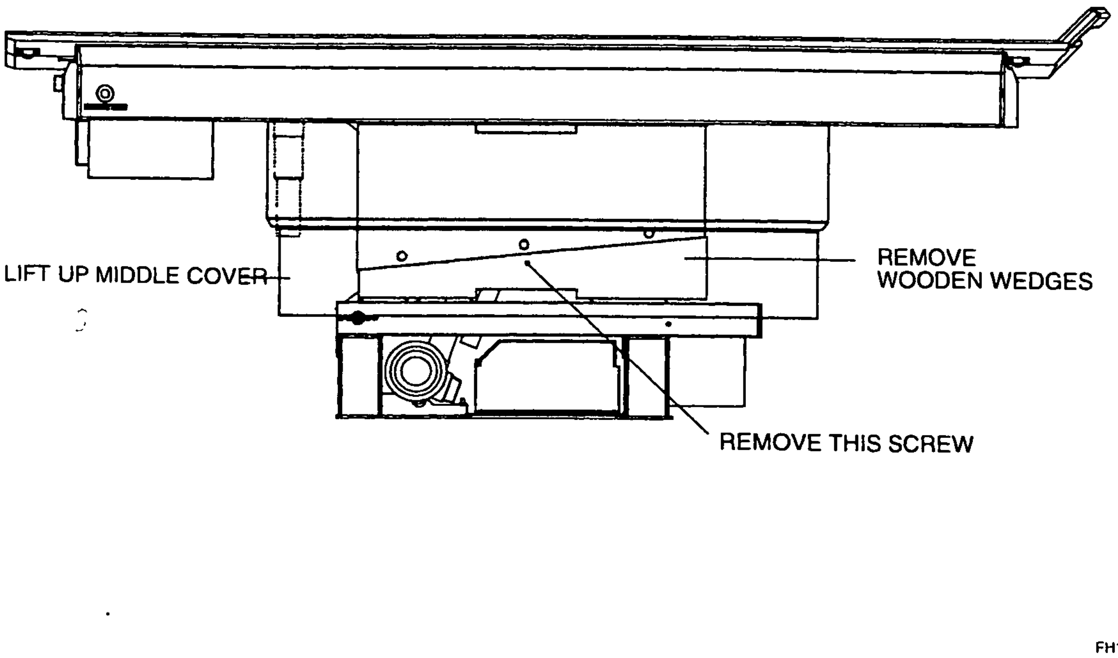
CAUTION

Vertical activation of the patient support, while the transport provisions are installed, will damage the patient support.

Remove the two transport provisions (left and right side) as follows.

1. With an electrical drill unscrew the screws holding the carton box. Remove the carton box.
2. Remove the box with the lower covers, the bag with the insulation spacers and mounting material, and the wooden alignment bars. Store these items in the magnet room. They will be used during the installation of the patient support.
3. Lift up the middle covers. It might be needed to put something underneath the middle covers, this to prevent it from falling down.
4. Take a Phillips screwdriver and a hammer, these are the tools needed to remove the transport provisions.
5. Remove the screws that lock the upper and lower wooden wedges.
6. Slide the lower wooden wedge in lengthwise direction out of the patient support. If needed use the hammer.
7. Remove also the upper wooden wedge.

Figure 27 - Remove transport provisions



Remove the screws holding the patient support table on the crate:

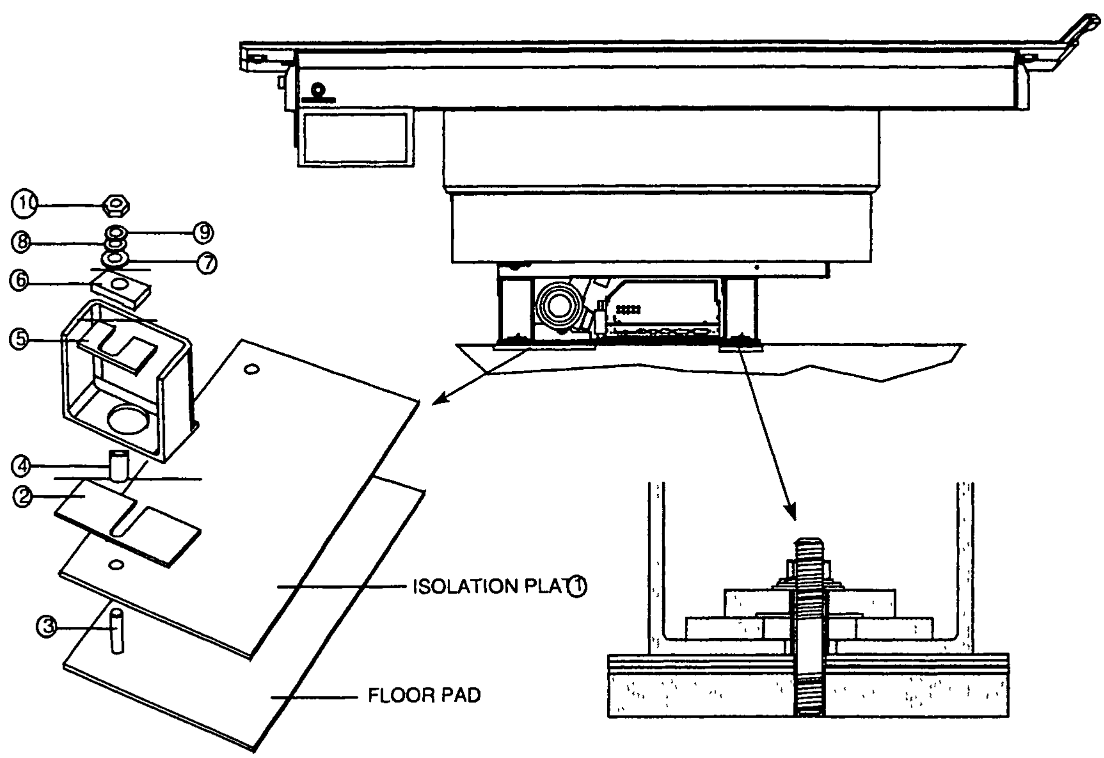
1. Take two 17-mm open-ended spanners, which are needed to remove the screws.
2. Loosen the four screws and nuts, which are used to fixate the patient support table to the crate. Remove the four screws and nuts.
3. Lift the patient support table from the crate and store it in the magnet room, where can be installed when magnet installation is done.
4. Remove the crate and other packing material out of the magnet room. Instruct the movers to take this packing material away from the site.

12.2. MOUNT THE INTERA PATIENT SUPPORT

For the mountıng procedure of the patıent support use drawıng Z3-5.1.2 together wıth the next procedure.

1. Posıtıon the large ısolation plate on the pad closest to the magnet.

Figure 28 - Posıtıoning the Intera patient support



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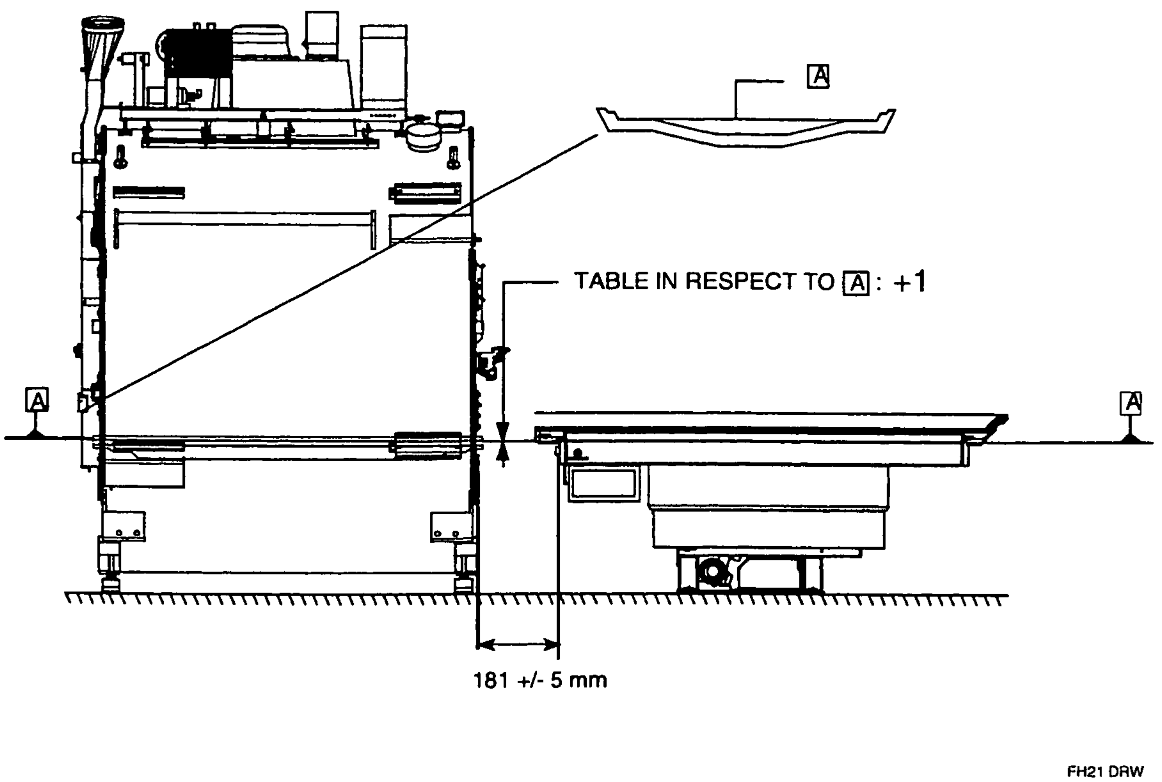
2. Posıtıon the patıent support on top of the floor pads, and the large isolatıon plate (1).
3. Insert one ısolation spacer (2) between front patıent support frame and floor pad.
4. Mount the four threaded rods (3).
5. Mount the ınsulatıon tubes (4) over the threaded rods (3).
6. Mount the fıxatıon plate (5) (10 mm thıck).
7. Mount fıxatıon block (6). The hole ın the fixation block is not ın the center. Thıs ıs done to enable mountıng of thıs fıxatıon block wıthin the "Ū" of the u-bar legs. The lower covers will be mounted tightly agaınst these vertical u-bars. Therefor the patient support table mounting material must be ınstalled wıthın the u-bar.
8. Mount ısolatıon washer (7), washer (8), lock washer (9) and nut (10). Do not tıghten the nuts yet.

The "travel-to-scanplane" is a fixed value, therefor the patient support table must be positioned on a given distance from the magnet:

Check/adjust the distance between the patient table and magnet gradient carrier to 181 mm +/- 5 mm. This distance is mainly determined during the magnet installation. When the template was used and the magnet has been positioned carefully the distance should be correct. Depending on the accuracy of the patient pads (screw holes) as done by the RF cage supplier, some adjustment is possible by moving the patient support.

Figure 29 - Position patient table



12.3. LEVEL THE INTERA PATIENT SUPPORT

12.3.1. INTRODUCTION

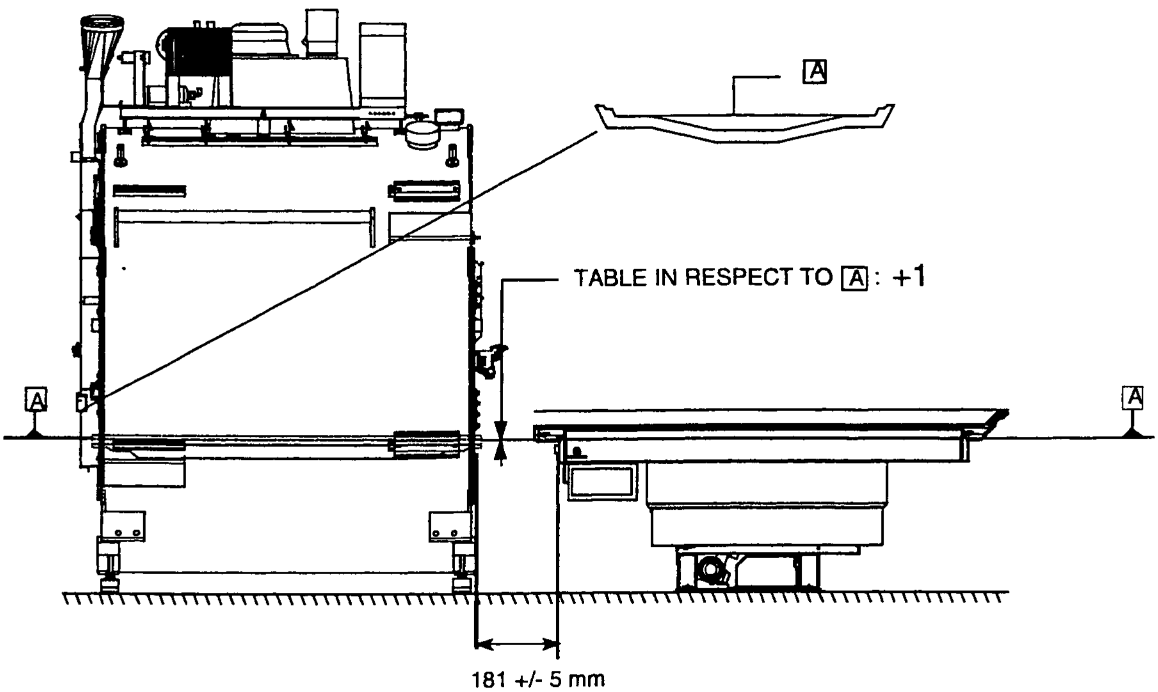
To level the patient table, the bridge section must be installed Do this as follows:

- Position the “bridge section” with the “finger protection switch plate” towards the patient support.
- Position the four interface points of the “bridge section” above the four holes in the magnet covers.
- Lower the “bridge section ” and push it down.

The patient support Intera is delivered in its upper operating position. It is not needed to move the patient support electrically upwards before the height adjustment. Both adjustments can be made without patient support cables connected.

Plane "A" is the running surface of the inner bed. The running surface of the patient support must be 1 mm above the running surface "A" of the inner bed when there is no load on the table.

Figure 30 - Position patient table



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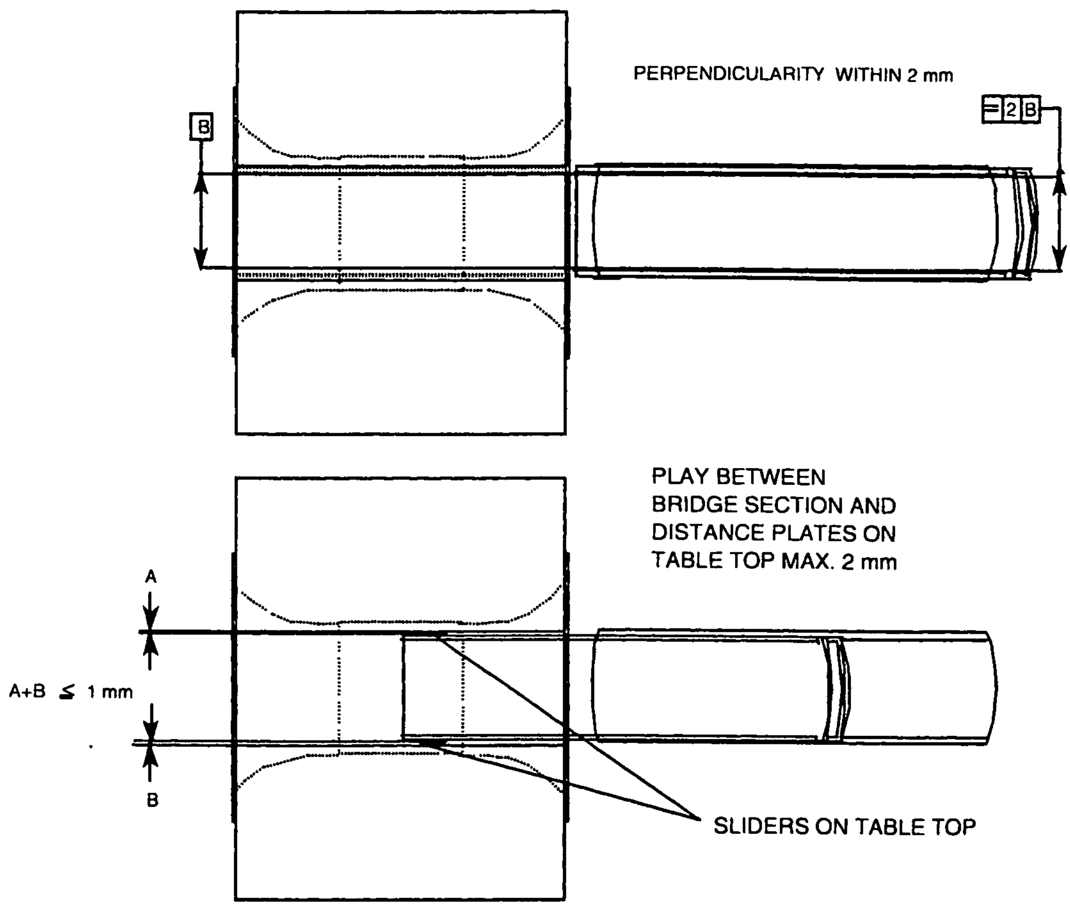
Perpendicular alignment between patient support and bridge section must be less than 2 mm.
See adjustment procedure on the next pages.

Figure 31- Perpendicular table adjustment



To achieve both the height and the perpendicular alignment perform the following actions:

12.4. ADJUSTMENT OF THE TABLE HEIGHT

1. Levelıng the patıent table:

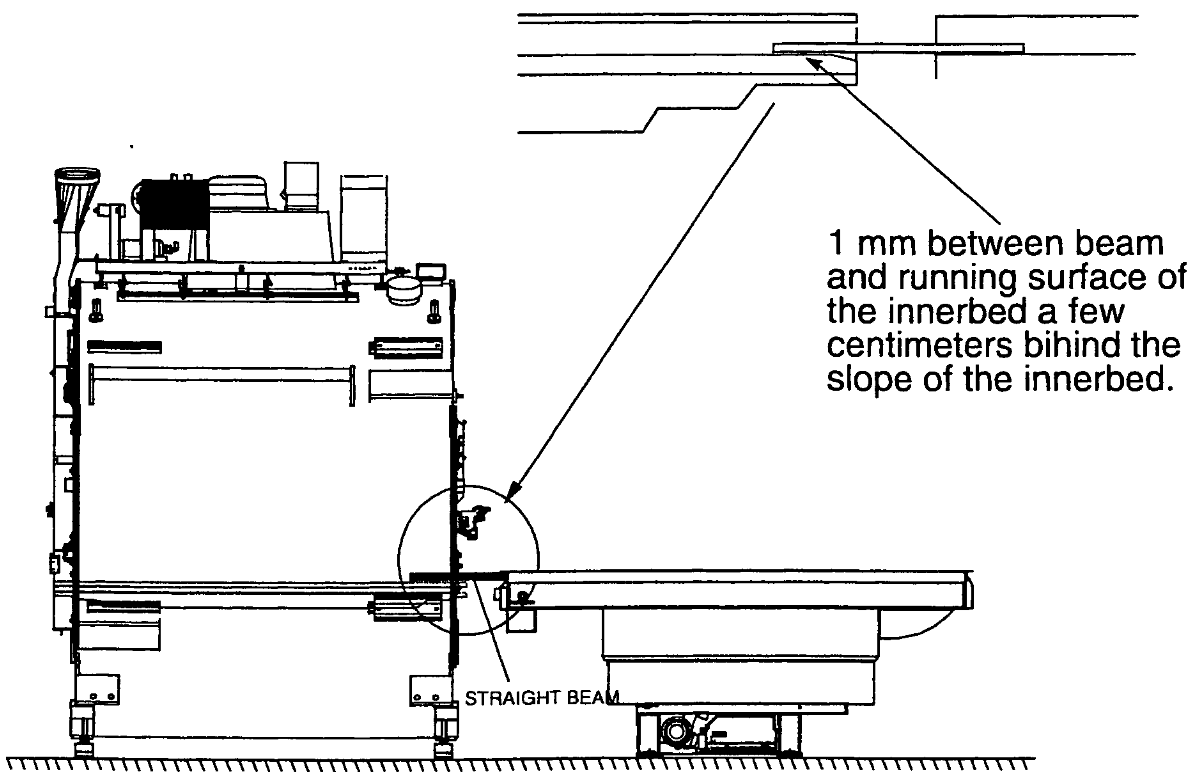
- Level the patıent table ın horizontally both longways dırection and perpendıcular on the longways dırectıon usıng the spırıt level. If needed add a spacer to level the patıent table.

2. The table heıght adjustment·

   - Mount the brıdge sectıon
     - Posıtıon the "bridge sectıon" wıth the "fınger protectıon swıtch plate" towards the patıent support.
     - Posıtıon the four ınterface poınts of the "brıdge sectıon" above the four holes ın the magnet covers.
     - Lower the "bridge sectıon " and push it down

   - Place the two yellow beams (supplıed wıth the patıent support) on top of the runnıng surface of the patıent support together wıth a spırıt level. Adjust the heıght of the patıent table (usıng ınsulatıon spacers below the patıent table frame) so that there is a 1 mm gap (use a 1 mm spacer) between the beams and the runnıng surface of the bridge section measured a few centımeters behınd the slopıng surface of the brıdge sectıon.

Figure 32 - Adjust table height

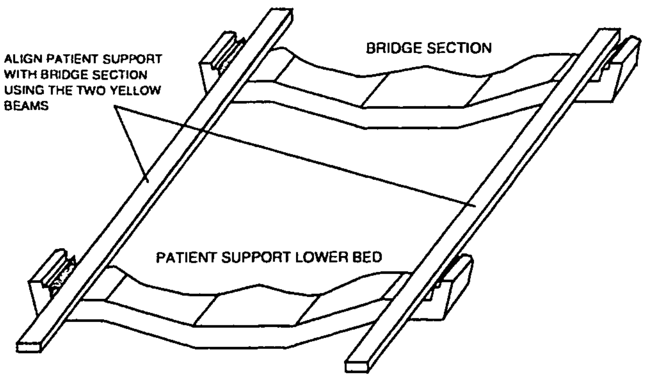


FH22 DRW

12.5. ADJUST THE PATIENT TABLE PERPENDICULARITY

1. The surfaces of the bridge section and patient support lower bed must be in one line (see next figure). Check (by using the two yellow beams) that these surfaces are in one line.

Figure 33 - Align patient support with bridge section

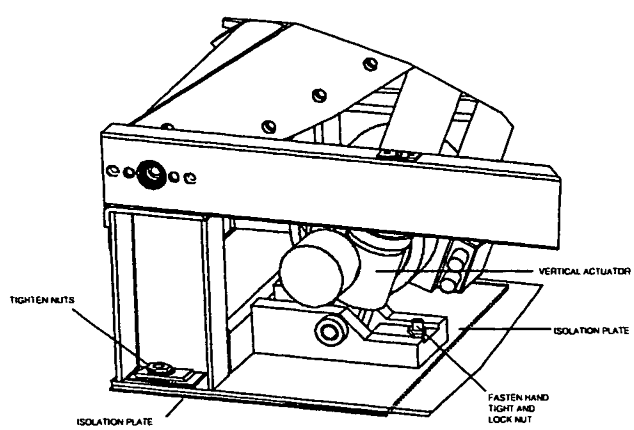


2. Fasten the patient support to the floor by tightening the four nuts.
3. Tighten (hand tight) the threaded rod located near the vertical actuator. In this way, the threaded rod pushes against the isolation plate on the floor so that the patient support frame can withstand downwards forces of the vertical actuator.
4. Lock the nut in order to lock the threaded rod.

CAUTION

Tighten the actuator rod, otherwise the frame can be damaged by placing loads.

Figure 34 - Fasten actuator nut



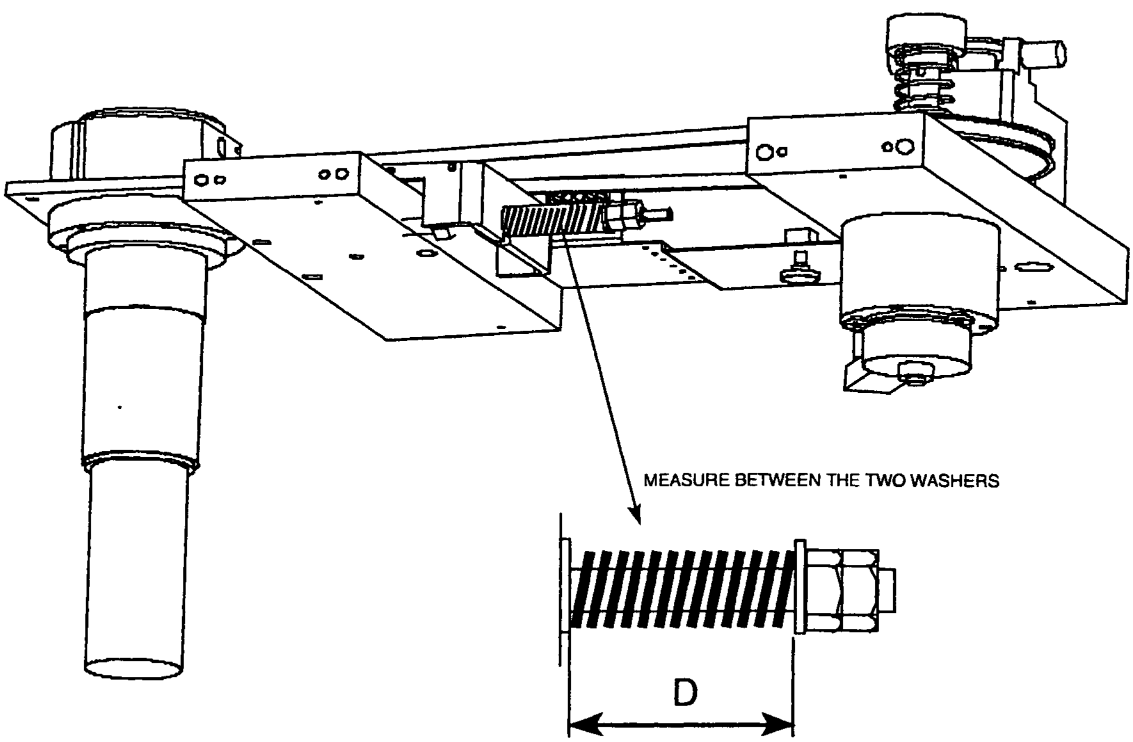
12.6. ADJUST HORIZONTAL TIMING BELT TENSION

The horizontal timing belt must be adjusted by compressing a spring.
Due to the magnet field, a force pulls to the horizontal motor assembly towards the magnet.
For different field strengths, the spring compression is different.

Procedure.

1. Remove the bottom cover underneath the table stretcher.

Figure 35 - Adjust timing belt



2. Adjust the distance D of the spring to:

   42 +/- 1 mm for 1.5 T
   52.5 +/- 1 mm for 1.0 T
   57 +/- 1 mm for 0.5 T (factory adjusted to this value)

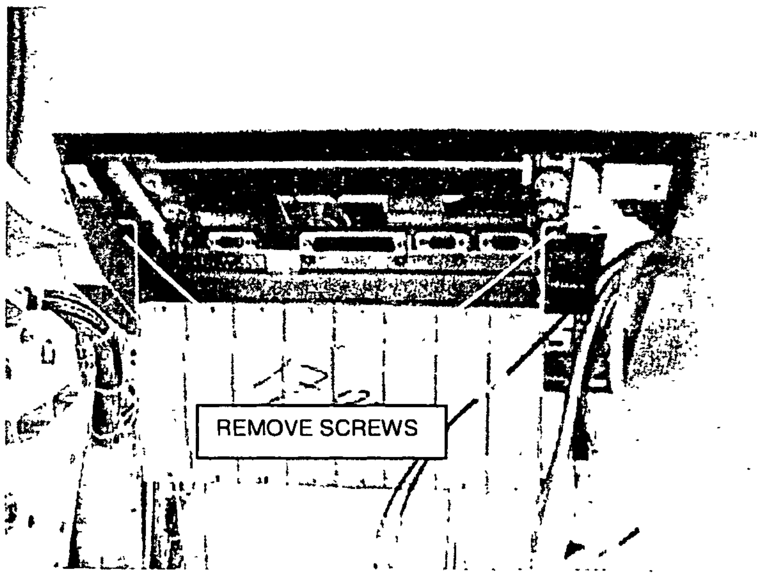
3. Lock the nut with the second nut.
4. Install the cover.

12.7. PHYSIOLOGY RIGHT OR LEFT SIDE

The Physiology unit is default installed at the left side front underneath the patient support.
The physiology unit must always be mounted at the same side as the PICU.
If the PICU is mounted on the right side onto the magnet, than mount the Physiology unit to the right side as follows.

1. Remove the physiology cover from the physiology unit.
2. Remove the lower front cover (magnet side).
3. Unscrew the two screws that hold the physiology unit to the patient support frame. The unit will rotate downwards so that you gain access to the cabling.
4. Disconnect all cables.

Figure 36 – Unscrew Physiology unit



5. Dismount the complete physiology unit and mount it at the right side underneath the patient support.
6. Route the cables to the right side and connect them.
7. Mount back the covers.

For mounting of the physiology units, refer to section 19 of this manual.

12.8. MOUNT AND ADJUST THE TABLE TOP

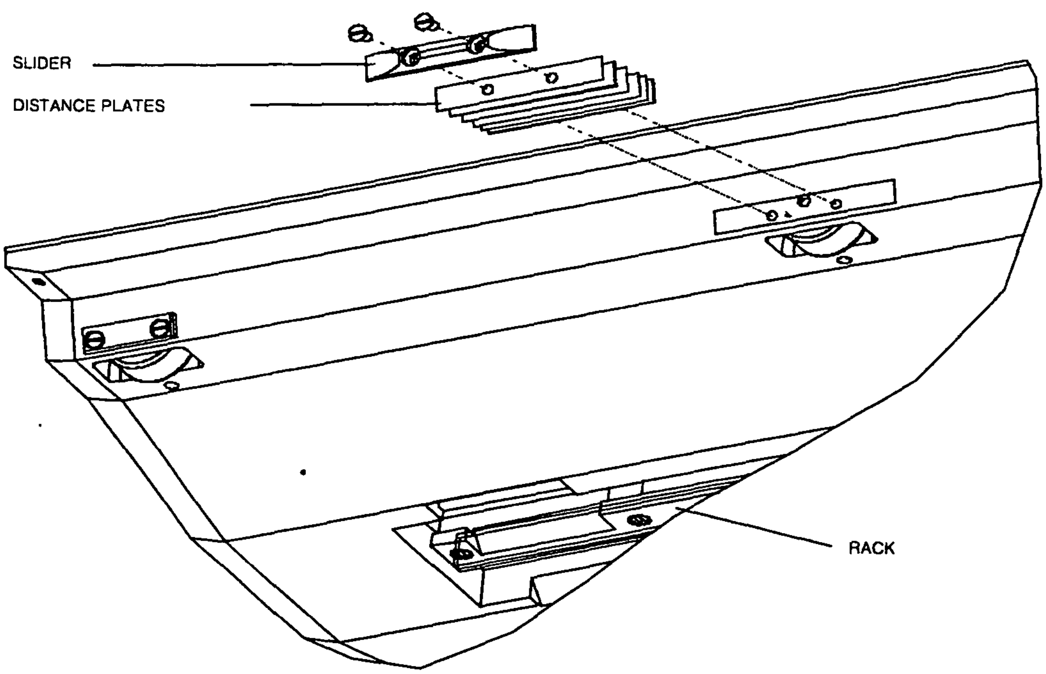
Adjust sideways play:

Maximal 2-mm play is allowed between the table-top and bridge section and between the table-top and table stretcher Minimize the sideways play as follows:

Check the side-ways play by moving the table-top manually completely in and outwards. Add or remove distance spacers until the table-top sliders are touching the bridge section or/and table stretcher. Than remove one spacer on the right and left side of the table-top.

It may not always be possible to add an equal amount of distance plates on right and left side. This is caused by the table-top position, which is not always symmetrical between the table stretcher.

Figure 37 - Adjust side ways play



Check the alignment without load, and with a load of ± 100-kg, by moving the tabletop in and out by hand. No bumping or scraping must be felt.

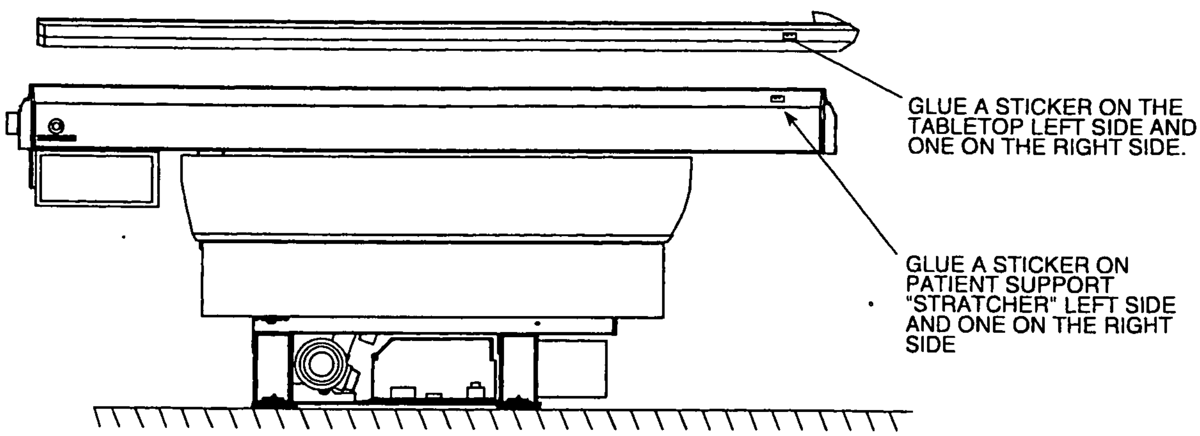
12.9. IDENTIFICATION STICKER

A tabletop Release 7 that is adjusted for a particular system cannot be used on another system
The yellow tabletop for Release 7 can be used on a trolley.

If a hospital has two or more systems near to each other, and a trolley option is used, an operator must be able to determine which tabletop belongs to which system. Identification stickers are therefor supplied with each tabletop. These stickers must be glued on the patient table and on each tabletop.

Glue one sticker (MR1) on the patient table of system 1. Glue another sticker [MR1] on a tabletop belonging to system 1. If there is more than one table top for system 1 then also glue a sticker [MR1] on those tabletops. Do the same for system 2 (sticker [MR2]) and system 3 (sticker [MR3]) if present.

Figure 38 - Location of identification sticker



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12.10. ONE TROLLEY FOR MORE SYSTEMS

If a trolley release <6.2 (4522 131 06814) or a trolley release 6.2 (4522 131 61701) is present in the hospital, than this trolley must be modified During installation **check if old trolleys are present** and if it is present, be sure that it is modified using the upgrade kit

WARNING

The table-top for Release 7 may not be used on an old type trolley. ***This creates an unsafe situation*** *that must be avoided. The table-top can roll on the trolley. There is a risk of fingers getting caught between handlebars of trolley and table-top. Be sure to update the old trolleys.*

In the commercial catalog, the customer is asked weather he has an old type trolley If the answer is YES, a modification kit will be send together with the initial system.

12.11. FINAL CHECK

1. Measure with an Ohmmeter that the patient support frame is electrically separated from the RF enclosure. You can measure between table frame and floor pads. The measured value must be more than 200 KOhm. Write down the measured value into the MRL.
2. Connect all cabling.
3. Mount the lower cover. Make sure to connect the safety grounds to these lower covers.

13. MOUNTING THE EXTERNAL COOLING SYSTEM TO THE TOP AIR DUCTS EXTENSIONS.

Mount the external air hoses (non-Philips) to the air ducts. Hoses must be fixed using hose clamping rings (non-Philips).

CAUTION

Take care that the air ducts and hose clamping rings do not vibrate against other objects like the false ceiling. This can cause image artefacts due to spikes.

14. MOUNTING THE PATIENT COOLING AIR DUCT

Mount the (delivered) patient cooling air duct between the patient cooling air duct at the magnet rear and the patient cooling unit inside the technical room.

Note that the air for the inlet of the patient cooling unit is taken out of the examination room. This is done to avoid disturbance of the air conditioning system on the examination room.